



Review

Sublimation crystallization: from mechanism to applications

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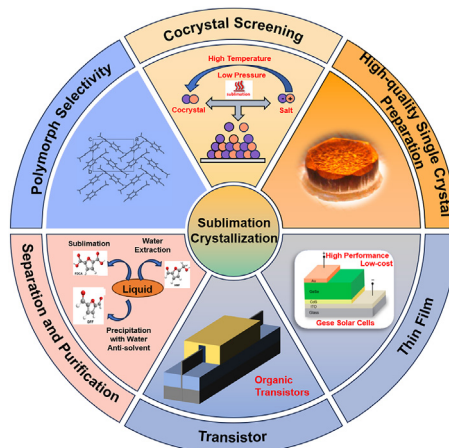
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HIGHLIGHTS

- The thermodynamic and kinetic mechanisms of sublimation crystallization were reviewed.
- Emphasis was placed on the separation and purification of substances crystallized by sublimation.
- The methods of selecting polycrystal and eutectic sublimation crystallization were summarized.
- The applications of sublimation crystallization in thin films and transistor materials were summarized.
- The prospects and challenges of sublimation crystallization for functional materials and purification were discussed.

GRAPHICAL ABSTRACT



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ABSTRACT

Sublimation crystallization, a solvent-free purification technique, has emerged as a sustainable strategy in materials science for producing high-purity substances and advanced functional materials. This review delves into the thermodynamic and kinetic principles governing sublimation crystallization, such as vapor pressure, temperature, and molecular interactions, which play crucial roles in phase transitions and crystal growth. We then explore its practical applications in the separation and purification of materials, the preparation of high-quality single crystals, polymorph and cocystal screening, and the fabrication of thin films and semiconductor devices. By clarifying the processes and mechanisms involved, this review aims to provide insights into how sublimation crystallization optimizes separation and purification techniques while enhancing material properties as a secondary benefit. This work also outlines future directions and challenges for refining high-efficiency purification methods and advancing sustainable chemical engineering technologies.

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1. Introduction

Sublimation crystallization, a venerable technique dating back to ancient times, involves the direct transition of a substance from a solid to a vapor phase and back to a solid without passing through the liquid phase [1–3]. Its origins can be traced to alchemy, where figures such as Zosimos of Panopolis in the first century CE employed sublimation to separate substances like mercury and sulfur [4]. These early uses demonstrated an intuitive understanding of phase transitions, predating modern thermodynamics. Over time, sublimation crystallization evolved from these foundational practices to become an essential method for purifying both organic and inorganic compounds, offering a unique approach to the separation and refinement of high-volatility materials [5, 6].

The formalization of sublimation crystallization as an effective method gained momentum during the 19th and 20th centuries, paralleling advancements in thermodynamics and materials science [7,8]. Researchers began to articulate the principles governing phase changes, vapor pressure equilibrium, and the molecular dynamics that drive the sublimation process [9–11]. The ability to produce high-purity crystals of substances such as iodine, camphor, and certain pharmaceuticals greatly contributed to its widespread industrial adoption [12]. The development of vacuum sublimation further advanced the technique, enabling materials to sublime at reduced temperatures, thus preserving the integrity of heat-sensitive substances [13,14].

In contemporary science, sublimation crystallization is undergoing a renaissance, fueled by progress in crystal engineering, material synthesis, and the separation and purification of complex compounds [15,16]. Enhanced control over crystallization parameters now permits the tailored design of crystal forms, including polymorphs, cocrystals, and salts, which are crucial for optimizing the properties of pharmaceuticals and functional materials [17]. Technological advances have also expanded the role of sublimation crystallization in producing high-quality single crystals, which are indispensable in crystallography for X-ray diffraction studies [18]. In the semiconductor industry, sublimation is instrumental for growing high-purity crystals of materials like silicon and gallium arsenide, which are vital for electronic devices [19]. Thus, sublimation crystallization continues to be a key technique not only for purification but also for the development of advanced materials where precise control over crystal structures and phase behaviors is essential.

This review begins by summarizing the thermodynamic and kinetic principles underlying sublimation crystallization, followed by a detailed exploration of its applications, such as substance purification, high-quality single crystal production, polymorph and cocrystal screening, and the improvement of thin-film and transistor performance. Finally, the review addresses the potential limitations of the technique and considers future developments, aiming to highlight the role of sublimation crystallization in advancing the field of high-purity and high-performance materials.

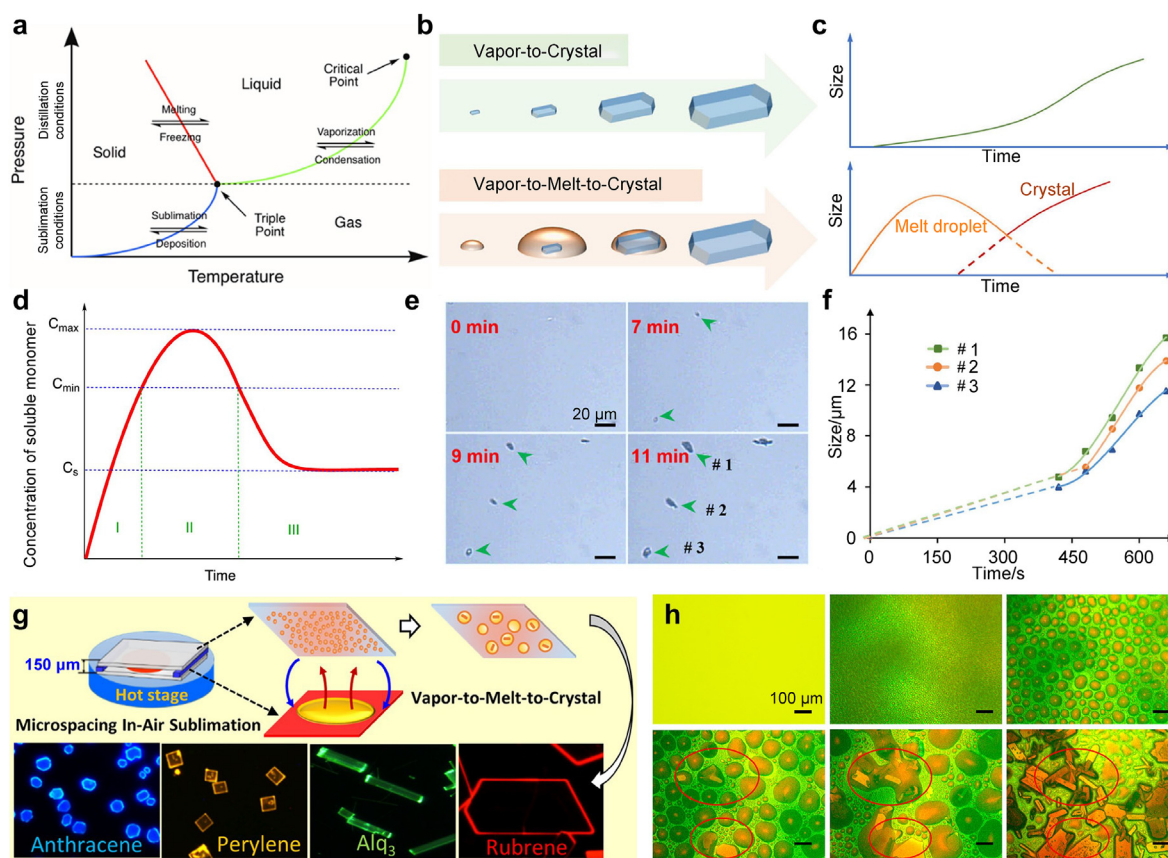


Fig. 1. (a) Phase diagram illustrating the physical states of the substance and the corresponding phase transition temperatures [36]. Copyright 2006 American Association for the Advancement of Science. (b) Schematic representation of the two mechanisms for crystal formation *via* sublimation: direct transition from vapor to crystal (top) and transition from vapor to melt followed by crystallization (bottom). (c) Variation in crystal size over time [46]. Copyright 2023 American Chemical Society. (d) Classical schematic of the LaMer model [47]. Copyright 2019 American Chemical Society. (e) Hot-stage microscopic images of PZA crystals over time, showing growth from the vapor phase on the substrate. (f) Measured size evolution of PZA crystals (#1–3) with time. (g) Microspacing in-air sublimation setup used for growing various organic single crystals on substrates. (h) Real-time observation of crystal formation on the top quartz substrate through hot-stage microscopy, showing the progression from melt droplets to crystals [48]. Copyright 2018 American Chemical Society.

2. Thermodynamics and kinetic mechanisms

The crystallization process involves phase transitions from atoms, ions, and molecules to colloids, nanocrystals, microcrystals, powders, films, or bulk single crystals [20–22]. This process is governed by both thermodynamic equilibrium and kinetic factors. Nucleation and crystal growth are the pivotal stages that determine the formation of various material forms [23–25]. In sublimation crystallization, the constituent elements transition directly from the vapor phase to the solid phase, bypassing the liquid phase, with the process being influenced by both thermodynamic driving forces and kinetic barriers. These interrelated processes operate across multiple scales, with thermodynamics dictating the equilibrium phase and kinetics controlling the rate and structure of phase transitions [26,27].

During the sublimation crystallization process, the law of energy conservation plays a dominant role. The total energy within the system, encompassing internal energy, kinetic energy, and potential energy, remains constant. Molecules that acquire sufficient energy exhibit a significant increase in their thermodynamic free energy. For a substance to undergo sublimation, the heat absorbed per unit mass must exceed the sublimation enthalpy (ΔH_{sub}) [28,29]. This is necessary to overcome the cohesive forces between the molecules in the solid state and the ambient pressure. At this stage, the molecules diffuse within the container's free space according to Brownian motion. If an environment capable of releasing the latent heat of phase transition (the heat released per unit mass during the phase change, equivalent to the enthalpy of sublimation) is created at the other end of the container, the deposition will occur, leading to the formation of crystal nuclei and subsequent crystal growth [30–32]. During sublimation, the system reaches solid-gas phase equilibrium, where the chemical potentials of both phases are equal. The driving force for sublimation crystallization arises from the chemical potential difference between the solid and gas phases, reflecting the tendency of the substance to transition from one phase to another [33]. This is analogous to supersaturation in solution crystallization and supercooling in melt crystallization, while in sublimation crystallization, it is indicated by the difference between the vapor pressure of the solid substance and the vapor pressure of its surroundings [34].

Phase diagrams are graphical representations that describe the state of a substance under various temperature and pressure conditions [35]. For sublimation processes, the focus is on the solid-gas equilibrium curve, also known as the sublimation curve (Fig. 1a) [36,37]. A typical phase diagram includes regions for solid, liquid, and gas phases, as well as the boundaries between these phases: solid-liquid, liquid-gas, and solid-gas. The sublimation curve indicates the conditions under which a substance transitions directly from a solid to a gaseous state. This curve is pivotal for understanding the sublimation process, as it outlines the necessary temperature and pressure conditions. In a phase diagram, the sublimation curve is situated between the solid and gas regions, extending from the triple point to areas of higher temperature and lower pressure. The sublimation point is the specific temperature and pressure at which a substance transitions directly from solid to gas [38]. On a phase diagram, this point is located on the solid-gas equilibrium line, representing a state where both solid and gas phases coexist [39]. For example, the sublimation point of carbon dioxide at 1 atm pressure is $-78.5\text{ }^{\circ}\text{C}$ [40]. Moreover, Du et al. [41] synthesized and separated transition metal sulfides by utilizing the temperature-vapor pressure relationship. Precisely determining the sublimation point is crucial for understanding a substance's thermodynamic properties and phase behavior.

The Clausius-Clapeyron equation can be employed to calculate the slope of the sublimation curve, thus determining the sublimation conditions at various temperatures and pressures [42,43]. In practical applications, the sublimation curve data is vital for designing and optimizing sublimation processes. For instance, in the purification of semiconductor materials, adjusting the sublimation temperature and pressure enables efficient separation and purification [44]. This application highlights the practical significance of understanding sublimation

dynamics. Advanced techniques such as Thermogravimetric Analysis (TGA) and Differential Scanning Calorimetry (DSC) are often employed to experimentally determine sublimation points and enthalpies. These methods provide precise measurements of changes in mass and heat flow under varying temperatures and pressures, offering insights into the sublimation characteristics of different substances. Thermodynamic cycles, like the Carnot cycle, provide a robust framework for analyzing energy conversion efficiency during sublimation. The heat exchange and energy transfer processes involved in sublimation closely mirror those in the Carnot cycle, where energy is transferred reversibly from a high-temperature source to a low-temperature sink [45]. The efficiency of the Carnot cycle is determined by the temperature gradient of the system. Through such analysis, energy utilization in the sublimation crystallization process can be optimized, making these processes more efficient and cost-effective.

The kinetic mechanisms of sublimation crystallization can be broadly categorized into two main types (Fig. 1b–c) [46], significantly influenced by the apparatus and material systems used. For traditional Physical Vapor Transport (PVT) and vacuum evaporation equipment, molecular vapor directly transforms into solid crystals or thin films. This mechanism, known as “vapor-to-crystal”, involves the sublimation of the target compound's solid phase at high temperatures to provide vapor, which is then transported to a low-temperature region by inert gas flow or sudden volume changes during sublimation and condensation, known as the “Stefan flow” [49–51]. When the hot vapor reaches the low-temperature region and contacts the cold surface, condensation occurs, leading to local supersaturation. According to the LaMer model, when the concentration reaches a critical supersaturation level (C_{max}), instantaneous homogeneous “burst” nucleation occurs, followed by diffusion-limited crystal growth due to continuous vapor supply (Fig. 1d) [47,52,53]. Nucleation can occur either directly in the vapor phase or on the contacting cold surface. After nucleation, continuous growth ensues, resulting in an increase in crystal size over time [54]. This mechanism is well illustrated in Su et al.'s [46] study on the sublimation crystallization of pyrazinamide (PZA). As shown in Fig. 1e–f, under hot-stage microscopy, small crystals appear on the underside of the top substrate after heating the bottom substrate (containing PZA powder) for 7 min. With continued heating, a stable molecular vapor flow provides sufficient supersaturation to sustain continuous crystal growth (at 9 and 11 min). However, the process of crystal formation on the sublimation surface is reversible, with some molecules potentially returning to the gas phase after colliding with the surface.

Recently, Tao et al. [48] reported a novel mechanism for sublimation crystallization from vapor to crystal in the context of organic semiconductor molecules, specifically rubrene. This process, termed “vapor-to-melt-to-crystal”, involves a molten phase intermediate. In this mechanism, vapor molecules first liquefy into a melt intermediate on the top substrate, followed by nucleation within the droplets according to Ostwald ripening rules [55]. Crystal growth occurs at the expense of the melt droplets, characterized by an initial increase in droplet size, followed by a decrease, and eventual steady increase in crystal size. The process of droplet formation and subsequent crystal growth on the top quartz substrate is depicted in Fig. 1g. The typical setup, as shown in Fig. 1h, consists of a hot stage and two silicon wafer substrates separated by small glass spacers. Notably, the micro-spacing and induced temperature distribution are crucial for this unique sublimation crystallization mechanism. At larger distances ($> 4\text{ mm}$), the material deposits as a polycrystalline film, similar to traditional vacuum deposition where vapor directly converts to solid. At smaller distances ($\sim 2\text{ mm}$), dendritic crystals form through a combination of initial melt droplet formation and subsequent crystal growth. The small temperature difference under micro-spacing conditions favors the formation of the intermediate melt phase. Compared to direct vapor-to-crystal crystallization, this process lowers the nucleation energy barrier. This is supported by Volmer analysis, indicating that the involvement of the liquid phase reduces the nucleation barrier, making the melt intermediate pathway more

Table 1
Classification methods of sublimation crystallization based on various criteria.

Criteria	Method	Characteristics	Scope	Advantages/limitations
Operational conditions	Atmospheric pressure sublimation [59]	Simple operation and low cost	Crude purification of low-purity compounds	High environmental impact, low purity
	Vacuum sublimation [60]	Reduced thermal decomposition	Thermosensitive compounds	Purity enhancement, residue reduction
	Pressurized sublimation [61]	Pressure-controlled sublimation dynamics	High vapor pressure substances	Side reaction suppression, crystallization acceleration
	Zoned sublimation [62]	Stepwise heating, sublimation and component separation	Fine separation of complex mixtures	Component purity enhancement, high separation precision
	Photo-thermal sublimation [63]	Laser-directed sublimation heating	High-precision crystal fabrication	Localized heat control, yield improvement
Mass and heat transfer mechanisms	Low-temperature sublimation [64]	Pressure-controlled sublimation at ultralow temperatures	Volatile substance purification	Prevention of thermal degradation
	Thermal gradient sublimation [65]	Thermal gradient condensation crystallization	Large and high-quality crystal growth	High-quality crystal fabrication, complex conditions
	Physical vapor transport (PVT) [66]	Gas-mediated sublimation product transfer, gas-phase crystallization	Industrial-scale wide-bandgap semiconductor crystal growth	High cost, complex process
	Chemical vapor transport (CVT) [67]	Gas-phase precursor deposition for thin films/crystals	Inorganic materials	Versatile, controlled deposition with potential for scalability
External field assistance	Electric field-assisted sublimation [68]	Electric field-induced molecular motion	Dipole molecule or ion crystal growth	Crystal orientation and morphology control
	Magnetic field-assisted sublimation [69]	Magnetic field-directed molecular motion for oriented crystal growth	Ferromagnetic or magnetically responsive crystal growth	High magnetic responsiveness
	Ultrasound-assisted sublimation [70]	Ultrasonic energy-enhanced molecular diffusion	Rapid nanocrystal formation	Controlled crystal size, uniform distribution

favorable [56]. These mechanisms are also applicable to inorganic materials. For instance, Chithaiah's [57] work on the chemical vapor deposition (CVD) of MoS₂ nanotubes follows the “vapor-to-crystal” pathway, while Tan et al.'s [58] synthesis of InGaAs nanowires via CVD is consistent with the classic metal-catalyzed vapor-liquid-solid growth mechanism.

Furthermore, a detailed classification of sublimation crystallization based on various categorization criteria has been conducted, including previously mentioned PVT, as outlined in Table 1.

3. Applications of sublimation crystallization

3.1. Separation and purification

Sublimation technology presents significant advantages in separation and purification processes, particularly when contrasted with traditional methods. For example, solvent crystallization is a widely used purification technique, but it is often constrained by the solubility of the target compound and can leave behind residual solvents, potentially compromising product purity [71,72]. In contrast, sublimation is solvent-free, eliminating the risk of residual contamination and ensuring high-purity outcomes. Distillation, a classical method for liquid separation, typically requires high temperatures, posing challenges for heat-sensitive materials and consuming considerable energy [73,74]. Sublimation, which operates in the solid phase, is better suited for volatile and thermally stable substances, and it offers lower energy consumption compared to distillation. Extraction, another method, relies on the partitioning of compounds between two immiscible liquids and often demands large amounts of solvents [75]. This increases costs and environmental concerns related to solvent disposal. Traditional separation techniques always rely on solvents, but sublimation does not rely on any solvent and completely avoids these problems. This makes sublimation particularly effective when handling solvent-sensitive materials, avoiding potential adverse interactions.

High-purity magnesium plays a pivotal role in industries such as automotive, aerospace, and electronics, as well as in the development of magnesium-based materials, owing to its outstanding properties, including low density, high electromagnetic shielding, superior thermal conductivity, excellent electrical conductivity, and remarkable stability [76]. The performance of magnesium is directly influenced by its purity, and enhancing this purity is key to further improving material properties.

However, magnesium's high reactivity and poor corrosion resistance present challenges [77,78]. Studies have shown that metallic impurities accelerate the corrosion of magnesium due to their higher hydrogen reduction exchange current density, which restricts the efficiency of magnesium production processes [79,80]. Currently, the most common methods for commercial production of high-purity magnesium include electrolysis and thermal methods, such as vacuum carbothermal reduction and the Pidgeon process [81,82]. In addition, Xiong et al. [83] demonstrated that vacuum sublimation technology can further enhance the purity of commercial magnesium (as shown in Fig. 2a). By comparing the saturated vapor pressures of different metals as a function of temperature (Fig. 2b), it is evident that the sublimation sequence follows the order: Zn > Mg > Ca > Pb > Mn > Al > Cu > Fe > Ni. Under specific conditions, elements with higher vapor pressures, such as Zn and Mg, are vaporized first. At 1023 K over 1.5 h, Mg achieved a 96% evaporation rate, with a recovery rate of 95.6%, representing optimal conditions. Adjusting the experimental parameters resulted in varying impurity removal efficiencies, as shown in Fig. 2c. Zn proved to be the most difficult to eliminate due to its similar vapor pressure to magnesium, achieving only a 41.7% removal rate. However, other impurities, such as Cu and Ni, were effectively removed, with their concentrations falling below detectable limits. These results highlight the effectiveness of vacuum sublimation in significantly improving the purity of magnesium, offering a promising method for industrial-scale material purification.

Zuo and colleagues [84] conducted a comprehensive study on the purification of arsenic, utilizing sublimation to achieve separation from antimony due to their distinct sublimation temperature ranges, 873.15–1173.15 K for arsenic and 904.15–1948.15 K for antimony (Fig. 2d). This separation was facilitated by the significant difference in their vapor pressures, allowing effective differentiation at temperatures below the melting point of antimony (904.15 K). Experimental results demonstrated that factors such as the structure of the condenser, temperature, and pressure had a pronounced impact on the separation efficiency. Segmental analysis of the condenser revealed that as the sublimation distance increased, the antimony content decreased markedly to as low as 0.21% farther from the heat source (Fig. 2e). Temperature adjustments further highlighted this effect; as the vacuum sublimation temperature increased from 698 K to 798 K, the antimony content in the condensates rose significantly, indicating that lower temperatures suppressed the formation of antimony clusters and thus reduced its concentration. Pressure also played a critical role, as

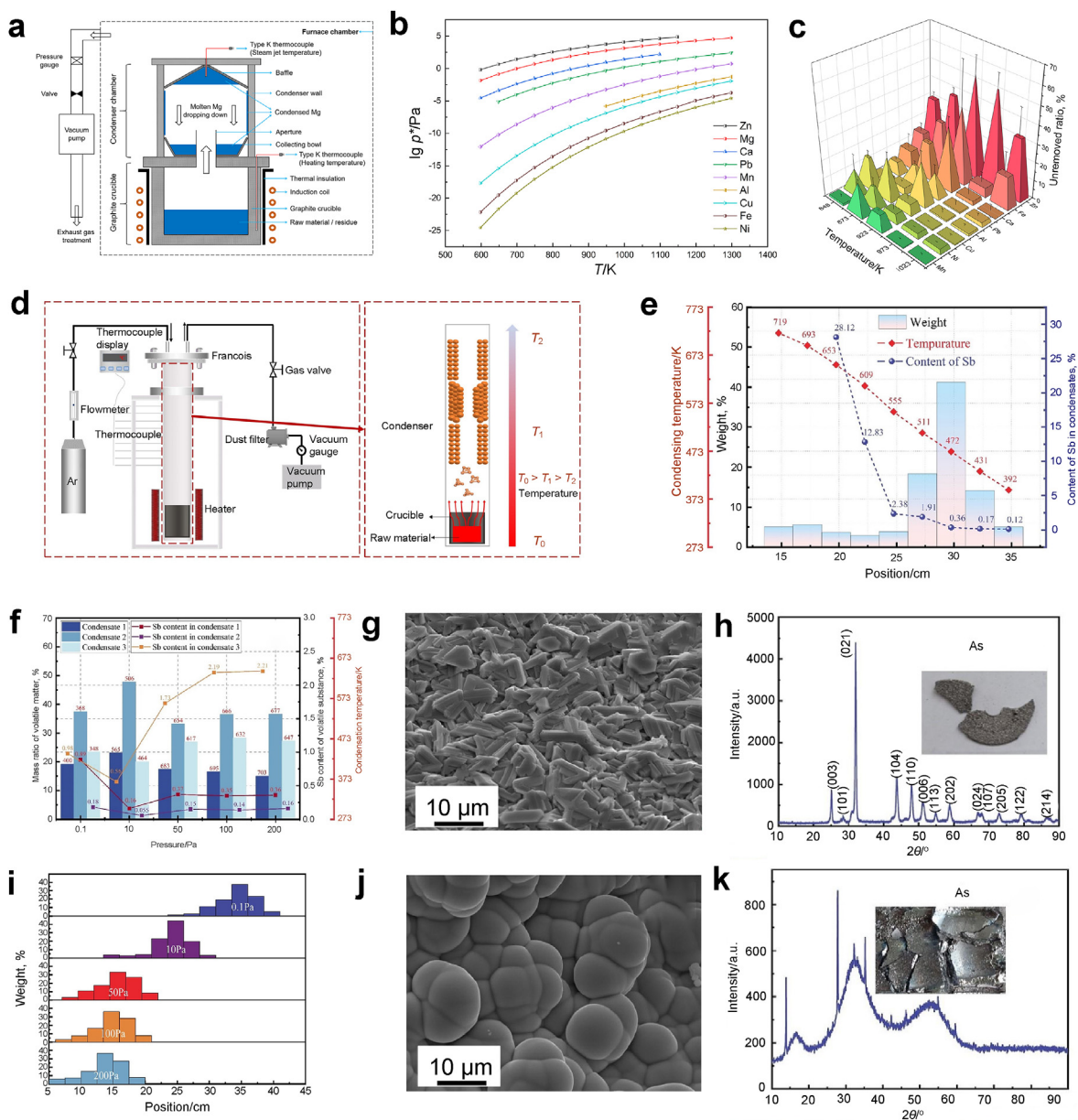


Fig. 2. (a) Cross-sectional schematic of the distillation apparatus. (b) Equilibrium vapor pressure of Zn, Mg, Ca, Pb, Mn, Al, Cu, Fe, and Ni in the temperature range of 600–1300 K. (c) Unremoved impurity ratios of Zn, Fe, Ca, Pb, Al, Cu, Ni, and Mn after vacuum distillation at heating temperatures of 848, 873, 923, 973, and 1023 K [83]. Copyright 2022 Springer Nature. (d) Structural diagram of the vertical vacuum sublimation furnace, illustrating raw material volatilization and condensation processes. (e) Experimental results under the conditions of 773 K temperature, 60 min holding time, and 10 Pa pressure. (f) Influence of pressure (at 748 K and 60 min holding time) on the efficiency of arsenic volatilization and antimony removal. (g) SEM image of condensate products near the heating zone's proximal ends. (h) XRD patterns of condensate products near the heating zone's proximal ends. (i) Distribution of condensates within the condensation zone at pressures ranging from 0.1 to 200 Pa. (j) SEM image of condensate products near the heating zone's distal ends. (k) XRD patterns of condensate products near the heating zone's distal ends [84]. Copyright 2024 Elsevier.

increased pressure heightened the temperature required for the gas-solid phase transition, and the increased presence of argon molecules intensified collisions with arsenic, hindering its movement and limiting antimony's mobility (Fig. 2f and i) [85,86]. Analysis of the resulting products revealed distinct differences between proximal and distal condensates: the former were porous and had rough surfaces, with polycrystalline and stepped distributions (Fig. 2g–h), while the latter were smoother, brighter, more brittle, and amorphous (Fig. 2j–k), with higher purity.

Sublimation technology has shown exceptional potential not only in the purification of metal materials but also in enhancing the performance of organic photovoltaics and organic semiconductors [87,88]. The design

and efficiency of organic photovoltaic cells are closely tied to exciton diffusion length (LD). Research by Curtin and colleagues [89] demonstrated a nonlinear increase in LD with improvements in film purity. As the purity rose from 97.1% to 99.0%, the concentration of exciton-quenching impurities decreased, leading to a significant increase in LD. When the purity exceeded 99%, further reductions in impurity levels resulted in even greater improvements in LD, emphasizing the critical role of material purity in maximizing LD.

In summary, sublimation crystallization presents itself as a highly efficient and green purification method, offering significant potential for preparing volatile and thermally stable materials. Compared to

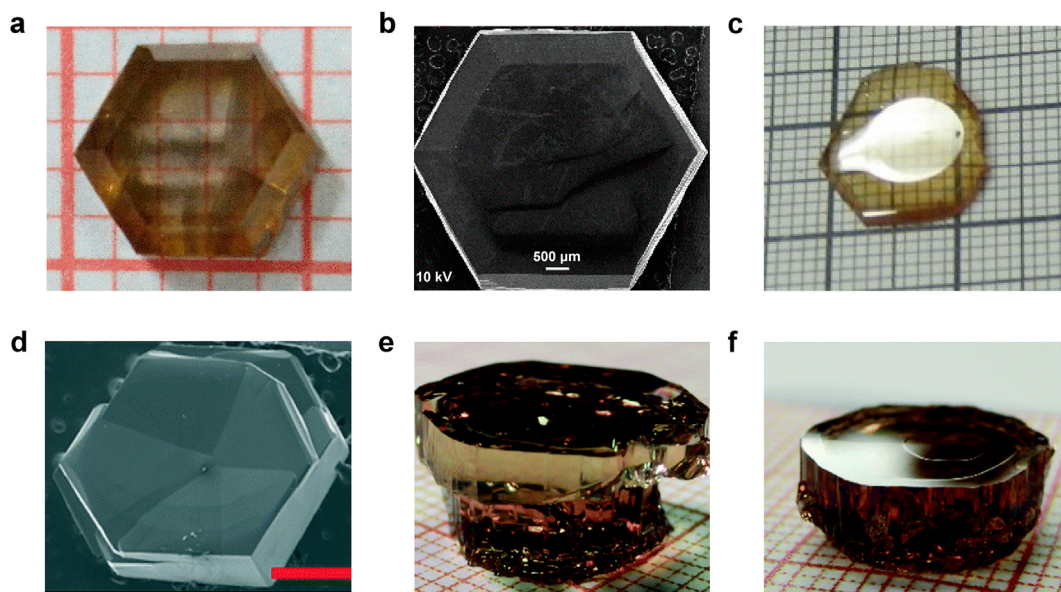


Fig. 3. (a) Optical image of the AlN crystal, with a grid size of $1 \times 1 \text{ mm}^2$. (b) Low-magnification SEM image of the bulk crystal, with a scale bar of $500 \mu\text{m}$, illustrating the overall crystal morphology [91]. Copyright 2019 MDPI. (c) Top view of a boule in reflection mode, showing well-developed side facets and a c-facet in the center, indicating step-flow growth and high crystal quality. The grown crystals are free of cracks [93]. Copyright 2010 Elsevier. (d) Low-magnification SEM image of a typical AlN crystal with a hexagonal facade, with a scale bar of 1 mm [94]. Copyright 2017 The Royal Society of Chemistry. (e) Design of the seed holder and corresponding AlN single crystal (Crystal 1). (f) Design of the seed holder and AlN single crystal (Crystal 2) [95]. Copyright 2016 The Royal Society of Chemistry.

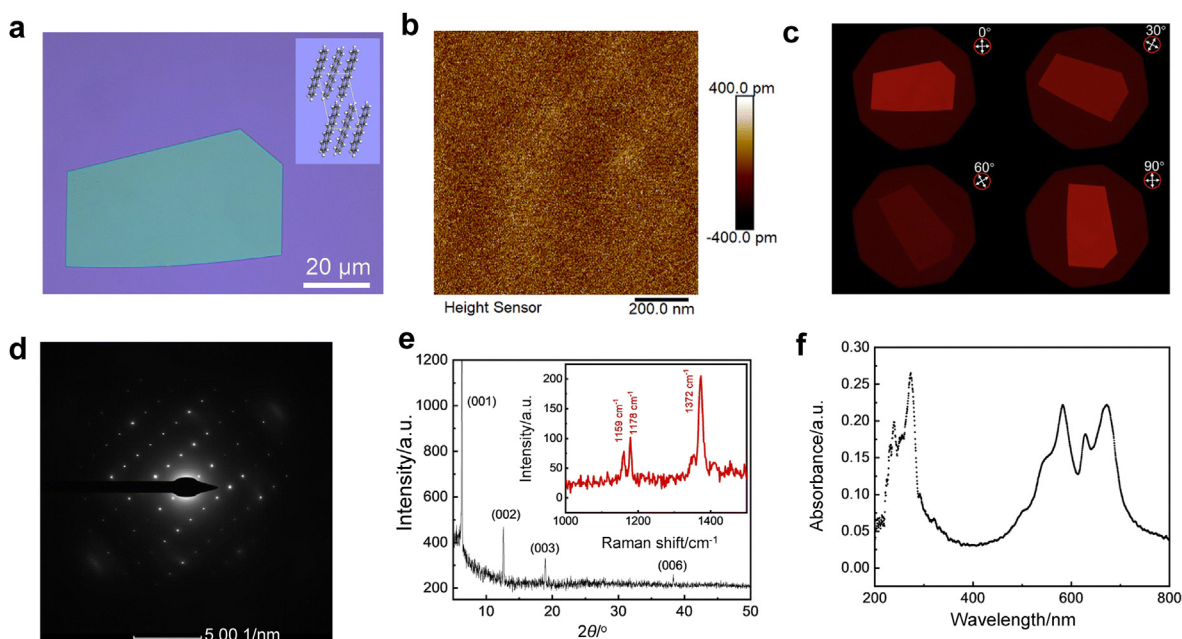


Fig. 4. Characterization of the pentacene single crystal: (a) Optical microscopy image of the pentacene single crystal. (b) Surface morphology of the pentacene single crystal. (c) Cross-polarized optical microscopy images under different polarization directions. (d) SAED pattern. (e) XRD pattern of the pentacene single crystal. Inset: Raman spectrum. (f) Light absorption spectrum of the pentacene single crystal on a quartz substrate [96]. Copyright 2023 The Royal Society of Chemistry.

conventional separation techniques, sublimation eliminates the need for solvents, thereby minimizing impurity introduction and secondary pollution while optimizing energy consumption. In areas such as metal purification, organic semiconductors, and photovoltaic technology, sublimation not only enhances material purity and performance but also contributes to the sustainable development of green chemical processes. As further advancements in process optimization and equipment technology emerge, sublimation crystallization is poised to play an increasingly critical role in the separation of complex systems and the

production of high-purity materials, solidifying its position as a key technology for future high-end material production.

3.2. High-quality single crystal preparation

Sublimation offers significant advantages in the production of high-quality single crystals. First, it allows for nearly complete utilization of the material, and second, it operates through physical phase transitions based on intrinsic material properties, such as sublimation temperature

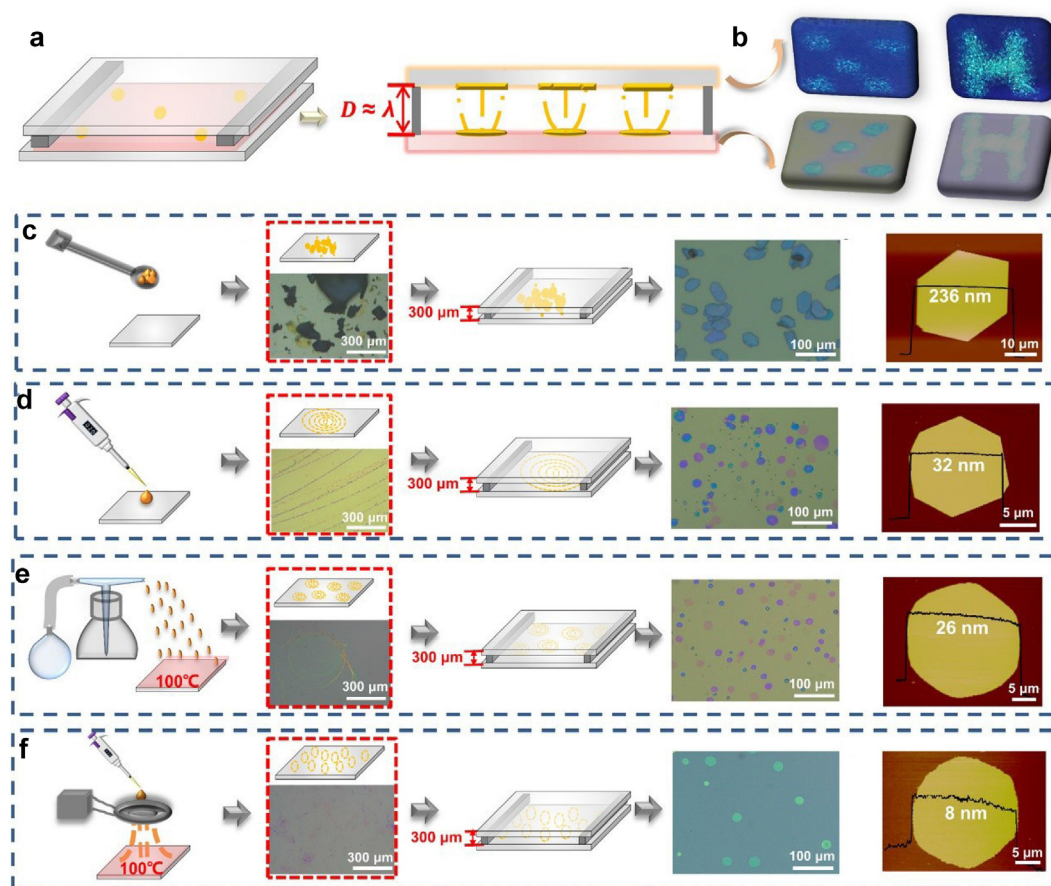


Fig. 5. Microspacing in-air sublimation (MAS) growth and deposition techniques for DNTT crystals: (a) MAS setup, showing two substrates separated by a spacing distance approximately equal to the mean free path (λ) of the vaporized molecules. (b) Photographs of the bottom substrates with patterned raw materials and the top substrates with as-grown crystals, which replicate the patterns. (c) Schematic illustrations and optical images of DNTT powders directly spread onto the substrate, showing different raw material distribution configurations. Optical microscopy images and AFM cross-section thickness profiles of DNTT single crystals grown under various conditions. (d) DNTT suspensions in toluene drop-cast onto the substrate. (e) DNTT suspensions sprayed onto the substrate using compressed gas. (f) Ultrasonic spraying of DNTT suspensions onto the substrate, demonstrating different deposition techniques [98]. Copyright 2020 American Chemical Society.

and vapor pressure, without the need for solvents. This results in a cleaner and more efficient process. For example, Mizuhara et al. [90] demonstrated that sublimation using a silicon carbide (SiC) substrate can produce high-quality aluminum nitride (AlN) crystals.

Chen et al. [91] further utilized a reverse temperature gradient method in the PVT process to grow high-quality amber-colored AlN crystals on tungsten substrates (Fig. 3a). Scanning electron microscopy (SEM) images showed excellent hexagonal symmetry (Fig. 3b), while a narrow full-width at half maximum (FWHM) of 66 arcseconds in the X-ray rocking curve confirmed the crystal's high crystallinity. Similarly, Herro et al. [93] grew AlN boules using PVT in a nitrogen atmosphere. The crystal initially grew as independent spheres, eventually forming a hexagonally symmetrical structure with a distinctive C-shaped facet (Fig. 3c), highlighting its superior quality. Liu et al. [94] achieved millimeter-scale AlN bulk single crystals using a dual-heater PVT method, producing highly ordered, hexagonally symmetric crystals with a maximum diameter of 22 mm (Fig. 3d). Moreover, Hartmann and colleagues [95] used the PVT method at 2050 °C to grow AlN crystals on N-polar AlN seeds. Effective control of thermal gradients (Fig. 3e–f) prevented parasitic growth, resulting in larger and more uniform crystals, demonstrating that fine-tuning thermodynamic conditions can yield even higher-quality crystals.

Organic single crystals have garnered significant attention due to their high carrier mobility and notable photoluminescence properties. Du et al. [96] developed stable pentacene single crystals using a simple air sublimation method, in which the initial material was sublimated at

260 °C for 5 min, followed by growth on a SiO₂/Si substrate, resulting in high-quality crystals (typically several hundred micrometers in length and about 72 nm in thickness) (Fig. 4a). atomic force microscopy (AFM) revealed the atomically smooth surface of the crystal (Fig. 4b). When observed through a cross-polarized optical microscope, the crystals exhibited changes in color and brightness upon rotation, indicating their highly ordered structure (Fig. 4c). Selected-area electron diffraction (SAED) patterns further confirmed their high crystallinity (Fig. 4d). X-ray diffraction (XRD) and Raman spectroscopy validated the composition of the crystals, while optical absorption spectra showed distinct peaks at 272, 583, 628, and 670 nm, suggesting potential applications in photo-detectors and light-responsive materials (Fig. 4e and f). Similarly, while 2-amino-5-nitrotoluene (MNA) is known for generating high-intensity THz radiation via infrared optical rectification, producing high-quality single crystals of MNA remains a significant challenge. By utilizing an initial sublimation step followed by slow solution crystallization, larger MNA crystals were grown, resulting in higher THz output, indicating the potential for enhanced THz generation with optimized crystal quality and size [97].

Further, Guo et al. [98] employed a close-distance sublimation method to grow dinaphtho[2,3-b:2',3'-f]thieno[3,2-b]thiophene (DNTT) single crystals, optimizing the raw material loading to achieve a minimum crystal thickness of 100 nm. The experimental setup was simple (Fig. 5a), with the distance between the substrate and raw material precisely calibrated to match the mean free path of DNTT molecules. This adjustment prevents vapor molecules from colliding and altering their

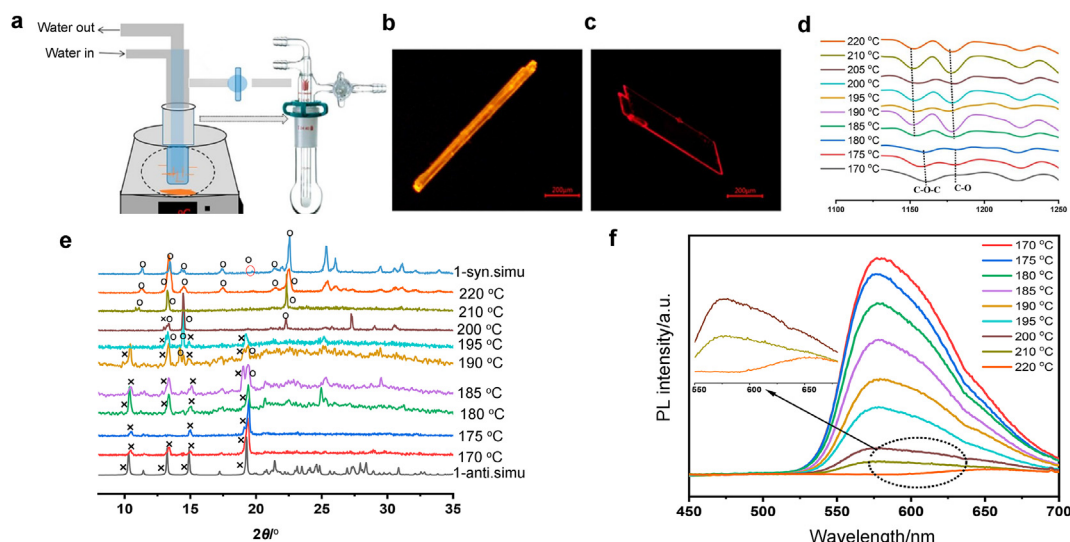


Fig. 6. (a) Schematic of the sublimation apparatus used for crystal growth. (b) Fluorescence microscopy image of polymorph A in the solid state under UV light excitation. (c) Fluorescence microscopy image of polymorph B in the solid state under UV light excitation. (d) C–O–C and C–O stretching vibrations of the sublimate occur at different temperatures, with dashed lines indicating changes in stretching patterns. (e) Powder X-ray diffraction (PXRD) patterns of the sublimate at various sublimation temperatures. (f) Photoluminescence spectra of the sublimate in the solid state at different temperatures ($\lambda_{\text{ex}} = 365 \text{ nm}$) [108]. Copyright 2018 American Chemical Society.

trajectory if the distance is too short, or from failing to reach the substrate when the distance is too far, thereby optimizing the yield. Interestingly, Guo's team [98] found a strong correlation between the properties of the raw material and the resulting crystals grown on the top substrate (Fig. 5b). Various preparation methods were explored, including conventional techniques, solvent-assisted deposition, compressed gas spraying, and ultrasonic spraying, each yielding raw materials with different sizes for sublimation. When ground raw material was sublimated at 230 °C, which is above the sublimation temperature but below the melting point, for 5 min, optical microscopy revealed large crystals with significant thickness, averaging over 400 nm (Fig. 5c). In contrast, solvent-assisted deposition using toluene as a dispersant resulted in smaller particles due to the coffee-ring effect, producing hollow ring-shaped materials. Sublimation under these conditions yielded DNTT crystals with a significantly reduced average thickness of 50 nm (Fig. 5d). Further attempts using compressed gas and ultrasonic spraying improved substrate material uniformity, mitigating the coffee-ring effect. Crystals grown using these techniques displayed thicknesses between 10 and 30 nm. The dispersion degree of raw material strongly influenced crystal growth during sublimation: better dispersion produced thinner and more uniformly contoured crystals (Fig. 5e–f).

Apart from a few organic materials, most inorganic compounds exhibit superior optoelectronic properties [99,100]. AlN, for instance, does not exist in nature as a mineral, making it crucial to explore new methods for producing high-quality single crystals. Traditional methods like solution-based or melt-based growth prove challenging for AlN crystal production, as nitrogen solubility in aluminum remains low, even under high-pressure nitrogen atmospheres. Consequently, these techniques are unsuitable for AlN crystal growth. In contrast, vapor-phase methods are more efficient, as the saturation vapor pressure of nitrogen in AlN is much lower under controlled conditions. Vapor-phase growth can be divided into physical and chemical processes. Among these, PVT has emerged as the most successful method for AlN crystal production, particularly when compared to hydride vapor-phase epitaxy, which is hindered by low growth rates and the use of toxic chemicals. PVT is not only suitable for AlN but also for other ultrawide bandgap materials, allowing for the production of large and high-quality crystals [101]. As crystallography continues to advance, more high-performance materials are being integrated into optoelectronic devices. The ability to

produce low-defect, thin, and compositionally uniform single crystals through vapor-phase growth substantially enhances the performance of these devices [102].

3.3. Polymorph selectivity

Crystal structures can vary significantly due to differences in molecular packing and conformations, resulting in distinct polymorphic forms, which is a common phenomenon during the preparation and storage of organic crystals [103,104]. These polymorphs, although derived from the same compound, can exhibit substantial differences in their physicochemical properties, such as solubility [105], melting and boiling points [106,107], sublimation temperature [108], decomposition temperature [109], and material performance [110–112]. Conventional methods for producing polymorphs, such as solvent evaporation [113], cooling crystallization [114], and mechanochemistry [115,116], are widely used due to their simplicity and ease of operation. However, the presence of solvents in these processes often leads to the inevitable inclusion of residual solvents in the final crystal product, and in some cases, results in the formation of solvates [117]. Solvates are multicomponent crystals formed when one or more solvent molecules are embedded in the crystal lattice in a specific stoichiometric ratio [118]. In the pharmaceutical industry, nearly one-third of organic compounds tend to form solvates, and these solvates may be toxic, unstable, or prone to decomposition, presenting challenges to polymorphic research [119].

Sublimation crystallization offers a distinct advantage as a modern technique for polymorph preparation. First, it eliminates the interference of solvent molecules, thus avoiding the formation of solvates or the inclusion of solvents in the final crystal structure which could complicate subsequent structural analysis. Second, sublimation allows for the efficient use of raw materials, maximizing the yield of polymorphs under controlled experimental conditions. Finally, by adjusting variables such as temperature, pressure, processing techniques, and the addition of seed crystals or templates, sublimation can control polymorphic transformations, making it a flexible and versatile method for polymorph production [120].

SiC exhibits multiple polymorphic structures, each of which significantly impacts the material's electrical and mechanical properties [121–123]. Yakimova [124] and colleagues demonstrated that at high

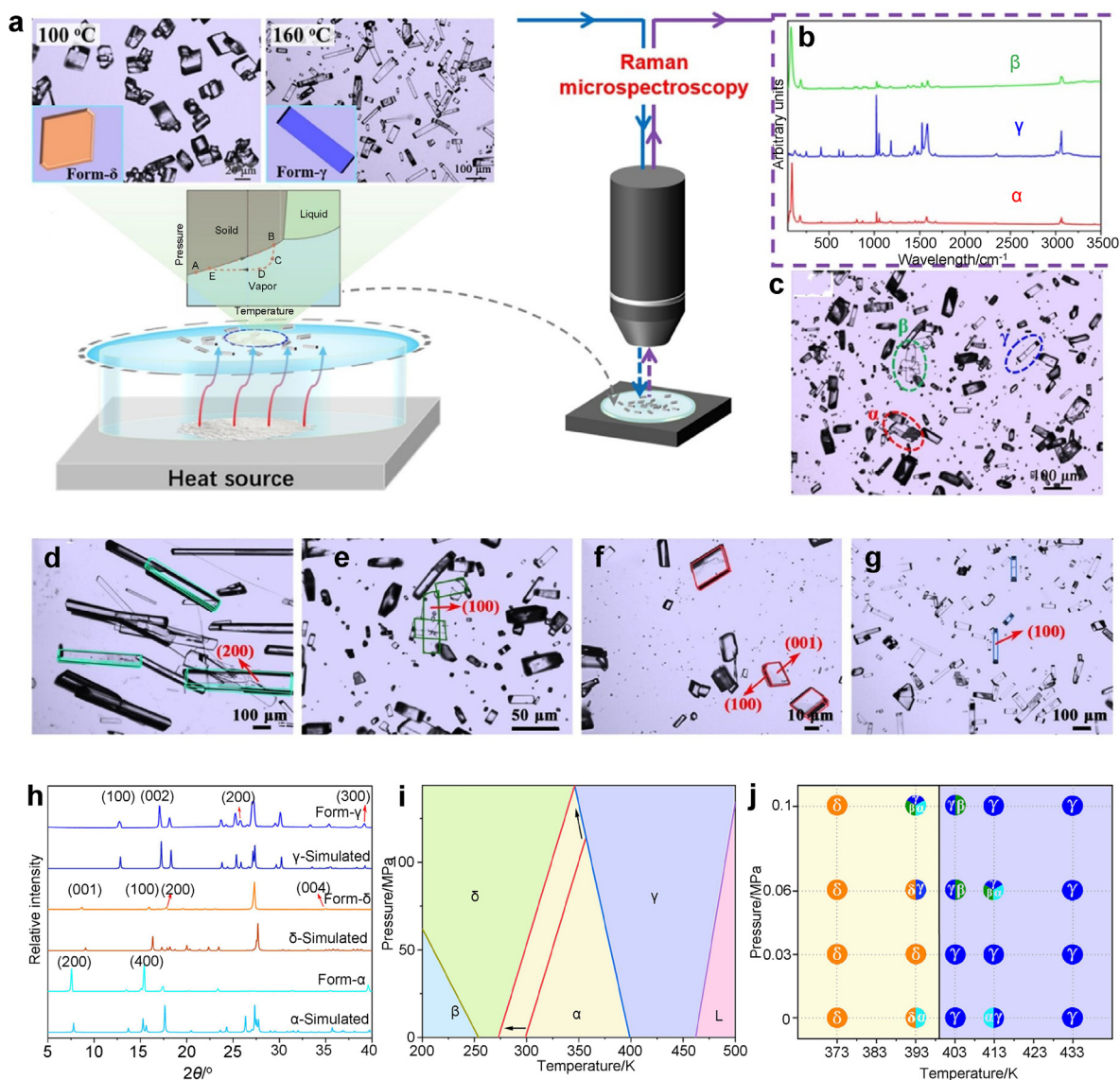


Fig. 7. (a) Schematic of the experimental setup and polarized optical microscopy images of the products at different temperatures. (b) Raman spectra corresponding to the α , β , and γ polymorphs of PZA. (c) Polarized optical microscopy image showing the α , β , and γ polymorphs of PZA. (d) Polarized optical image highlighting the PZA α polymorph (green box). (e) Polarized optical image highlighting the PZA β polymorph (green box). (f) Polarized optical image highlighting the PZA δ polymorph (red box). (g) Polarized optical image highlighting the PZA γ polymorph (blue box). (h) PXRD patterns of PZA polymorphs (α , δ , γ) and corresponding simulated patterns. (i) Pressure-temperature phase diagram of PZA and its five condensed phases. (j) Correlation between the selection of PZA polymorphs and sublimation temperature and pressure [46]. Copyright 2023 American Chemical Society.

temperatures (typically above 2000 °C), SiC powder can be sublimed, transforming into a gaseous state and then recondensing onto a seed crystal to form a new crystalline structure. At lower pressures, the gas phase density decreases, resulting in a slower crystal growth rate. This condition is favorable for producing certain high-temperature stable polymorphs, as the reduced gas density extends the growth time, allowing for greater atomic ordering. In contrast, at higher pressures, the gas phase density increases, making it easier for SiC gas to condense onto the seed crystal surface, promoting faster crystal growth. High-pressure conditions tend to favor the formation of low-temperature stable polymorphs, as the accelerated growth rate may facilitate the development of certain metastable crystalline forms. By precisely controlling the pressure within the system, it is possible to regulate the density of gas molecules and the mass transfer rate, thereby influencing the growth pathway of SiC crystals and enabling the selective formation of specific polymorphs. Similarly, carbamazepine, a pharmaceutical compound with five known

polymorphic forms, crystallizes as a mixture of Forms I and III when sublimed at 120 °C under a pressure of 0.2 mmHg [125].

By using conventional crystallization methods such as solution evaporation, cooling, liquid diffusion, and vapor diffusion, certain limitations arise, particularly when dealing with poorly soluble compounds or controlling polymorphic growth [126]. Sublimation crystallization, however, presents a more controlled approach, where selective crystallization of different polymorphs can be achieved by adjusting the sublimation temperature. For instance, Harada and colleagues [127] successfully obtained five polymorphs of 1-naphthyl derivatives via sublimation, while He et al. [19] produced polymorphs of 2,6-dichlorobenzene dicarboximide through similar methods. The ability to control the polymorphic forms of organic compounds using sublimation temperatures has been proven feasible. With applications to substances such as 1,1-bis(4-hydroxyphenyl)cyclohexane and 5-methoxysalicyloylhydrazine, the latter of which exists in two polymorphic forms [108].

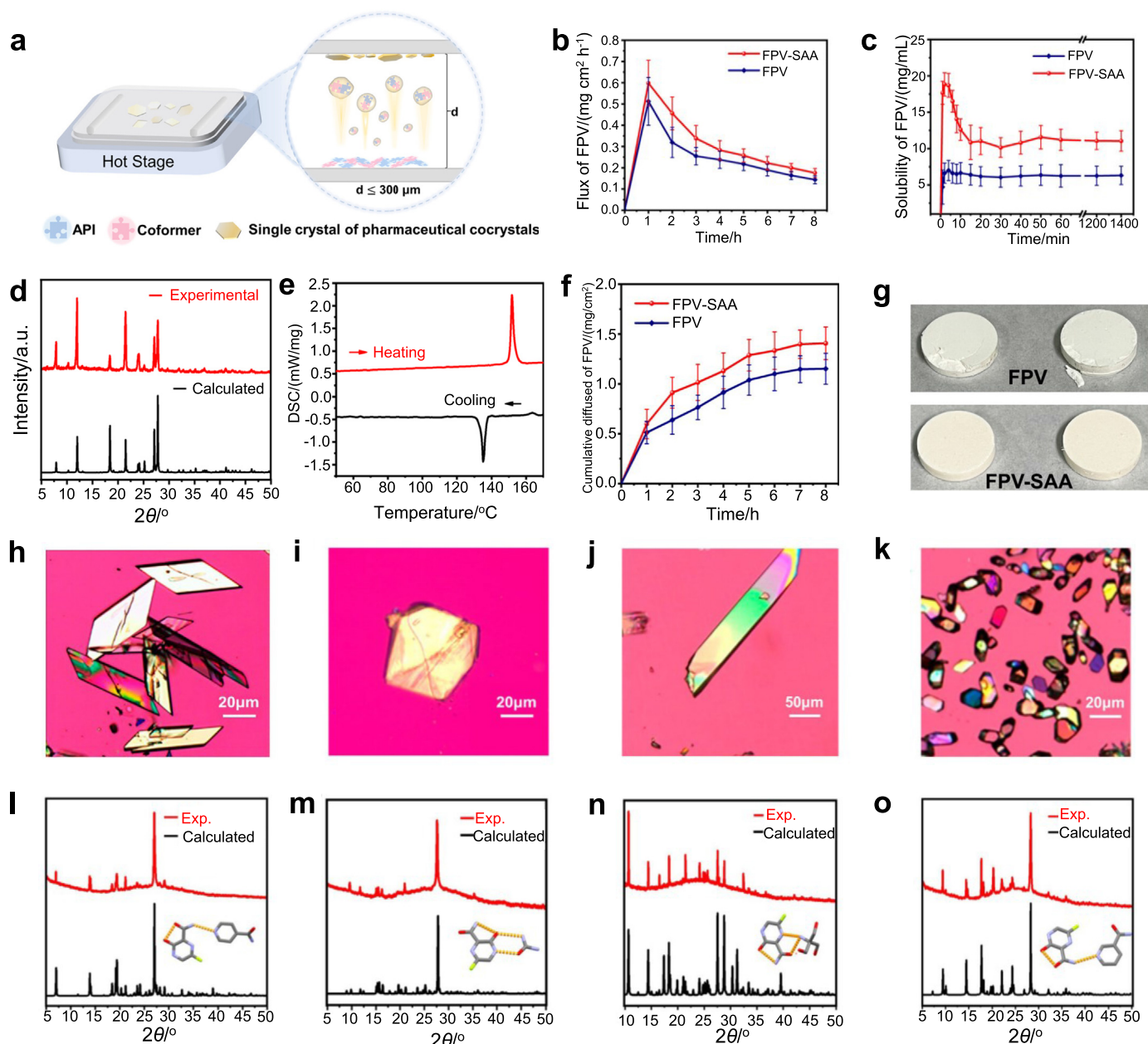


Fig. 8. (a) Schematic illustration of the MAS setup. (b) Flux measurements of FPV and FPV-SAA. (c) Equilibrium solubility curves of FPV and FPV-SAA. (d) PXRD patterns, showing experimental (red) and calculated (black) data. (e) Differential scanning calorimetry (DSC) curves with heating (red) and cooling (black) traces. (f) Cumulative release profiles of FPV and FPV-SAA. (g) Tablet images from tabletability studies, showing FPV (top) and FPV-SAA (bottom). (h–k) Microscope images of FPV-INIA, FPV-urea, FPV-Tro, and FPV-NIA. (l–o) PXRD patterns for the corresponding as-grown crystals. Inset: chemical structures of the cocrystals with color-coded atoms (gray for C, blue for N, red for O, and green for F). Peak intensities may be influenced by the substrate orientation used in the MAS method [141]. Copyright 2024 American Chemical Society.

Sublimation at 170 °C resulted in the metastable polymorph, while at 220 °C, the stable polymorph was achieved. Wang et al. [108] further explored the polymorphism of 5-methoxysalicylhydrazine through controlled sublimation temperatures (experimental setup shown in Fig. 6a). Adjusting the sublimation temperature resulted in the formation of two distinct polymorphs and a mixture of polymorphs (Fig. 6b–c). As the temperature rose further (180–220 °C), the intensity of these stretching bands increased, and they shifted toward 1145 cm⁻¹, indicating a conformational shift (Fig. 6d). PXRD analysis revealed a phase transformation as the sublimation temperature increased (170–220 °C, Fig. 6e). Furthermore, photoluminescence spectra showed a gradual decrease in fluorescence intensity at 560 nm as the temperature

increased (Fig. 6f). Infrared spectroscopy detected shifts in C–O–C and C–O stretching bands at 1161 and 1188 cm⁻¹ when the temperature was raised to 180 °C.

In another study, Su et al. [46] explored the crystallization process of pyrazinamide (PZA), flufenamic acid (FFA), and mefenamic acid (MFA) polymorphs using a micro-distance sublimation method (experimental setup shown in Fig. 7a). Taking PZA as an example, the team successfully obtained four distinct polymorphs of PZA by adjusting the temperature and pressure during sublimation (Fig. 7b–c), whereas conventional solution methods had only reproduced two polymorphs (α and δ). Microscopic analysis of the four PZA polymorphs obtained through micro-distance sublimation revealed distinct crystal morphologies: the

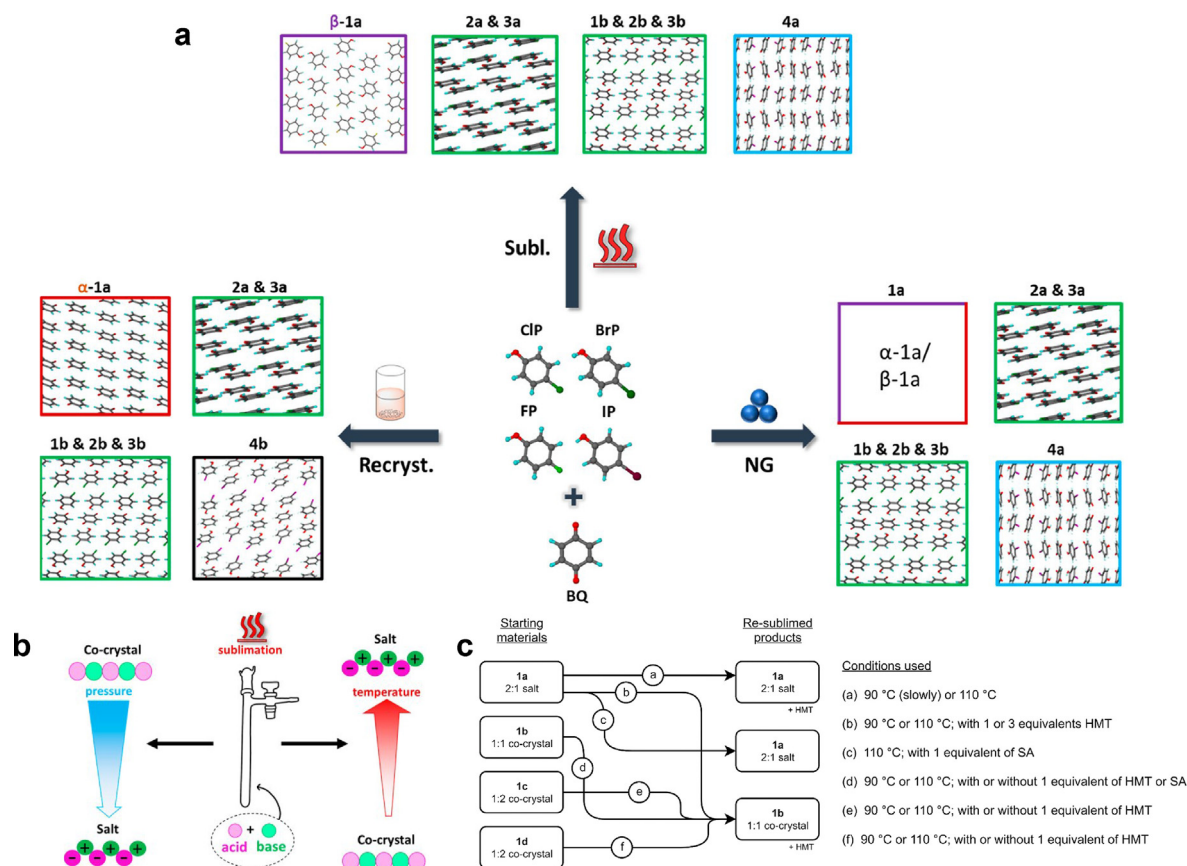


Fig. 9. (a) Summary of results from different crystallization methods for benzoquinone and *p*-halophenol cocrystals [143]. Copyright 2019 American Chemical Society. (b) Diagram showing sublimation outcomes for acid-base compounds with fixed stoichiometric ratios [144]. Copyright 2023 American Chemical Society. (c) Overview of the primary re-sublimation pathways for HMT and SA crystals. Each experiment lasted approximately 18 h, starting with 2 h under dynamic vacuum followed by a static vacuum, allowing for better separation of distinct product bands within the tube [145]. Copyright 2020 The Royal Society of Chemistry.

α form exhibited long rod-like shapes along the (200) plane (Fig. 7d), the β form showed thin rectangular plates along the (100) plane (Fig. 7e), the δ form presented a parallelogram-like structure along the (001) and (100) facets (Fig. 7f), and the γ form demonstrated rectangular prisms along the (100) plane (Fig. 7g). PXRD patterns further confirmed the polymorphic distinctions (Fig. 7h). Analysis of the PZA condensed-state *P-T* phase diagram suggested that temperature and pressure significantly influence the polymorphs produced; under vacuum conditions at 350–400 K, the α form is thermodynamically stable, while the γ form dominates at higher temperatures (Fig. 7i–j). Polymorph δ of PZA was found to be produced reproducibly at 373 K, whereas the α -form is the thermodynamic stable phase. Thereby, the polymorphic outcomes of PZA sublimation are predominated by crystallization kinetics. Overall, this study demonstrates that by controlling temperature and pressure, specific polymorphic forms of organic crystals can be selectively produced.

Lombard et al. [128] further demonstrated the potential of co-sublimation to produce polymorphs of cocrystal or salts, offering advantages over solution, melt, and mechanochemical methods. Comparative studies showed that co-sublimation generated polymorphs that are often unattainable through other methods. Their work highlighted that incorporating low molecular weight catalysts and crystal seed templates can enhance polymorph selectivity during vapor-phase crystallization. Kamali and colleagues [129] efficiently controlled carbamazepine (CBZ) polymorphism by introducing low molecular weight catalysts and complementary hydrogen-bonding agents at low concentrations, significantly lowering CBZ's sublimation temperature. These additives were not incorporated into the crystal lattice but facilitated selective polymorph crystallization. The mechanism involved the formation of a volatile

CBZ-acetamide hydrogen-bonding complex on the crystal surface, which delivered CBZ to growth sites at a lower temperature than CBZ dimer formation. Furthermore, adding crystal seeds allowed further control over the final polymorphic form. For instance, by attaching CBZ polymorph powder to the sublimation region with double-sided tape, they were able to crystallize all five CBZ polymorphs with high phase purity.

3.4. Cocrystal screening

Cocrystals are crystalline materials composed of two or more distinct molecular entities arranged in an orderly pattern within the same crystal lattice, stabilized through various intermolecular forces such as hydrogen bonding, π -stacking, or van der Waals interactions [130–132]. When proton transfer accompanies hydrogen bond formation, the multicomponent crystal is classified as a salt. Typically, these molecular entities consist of an active pharmaceutical ingredient (API) and a coformer [133]. Cocrystallization serves as an effective approach for modifying the physicochemical properties of drugs, including solubility [134], melting point [135], hygroscopicity [136], compressibility [137], and bioavailability [138]. The most common methods for cocrystal preparation include solvent-mediated crystallization [139], melt crystallization [140], and sublimation crystallization [128]. Among these, solvent-mediated crystallization is widely used and relies on factors such as solvent selection, solubility, evaporation rate, and crystallization temperature to control cocrystal formation. However, this method has limitations when dealing with poorly soluble, easily oxidized, or solvated compounds. Melt crystallization involves heating to a molten state followed by cooling to form the cocrystal, but this method often requires

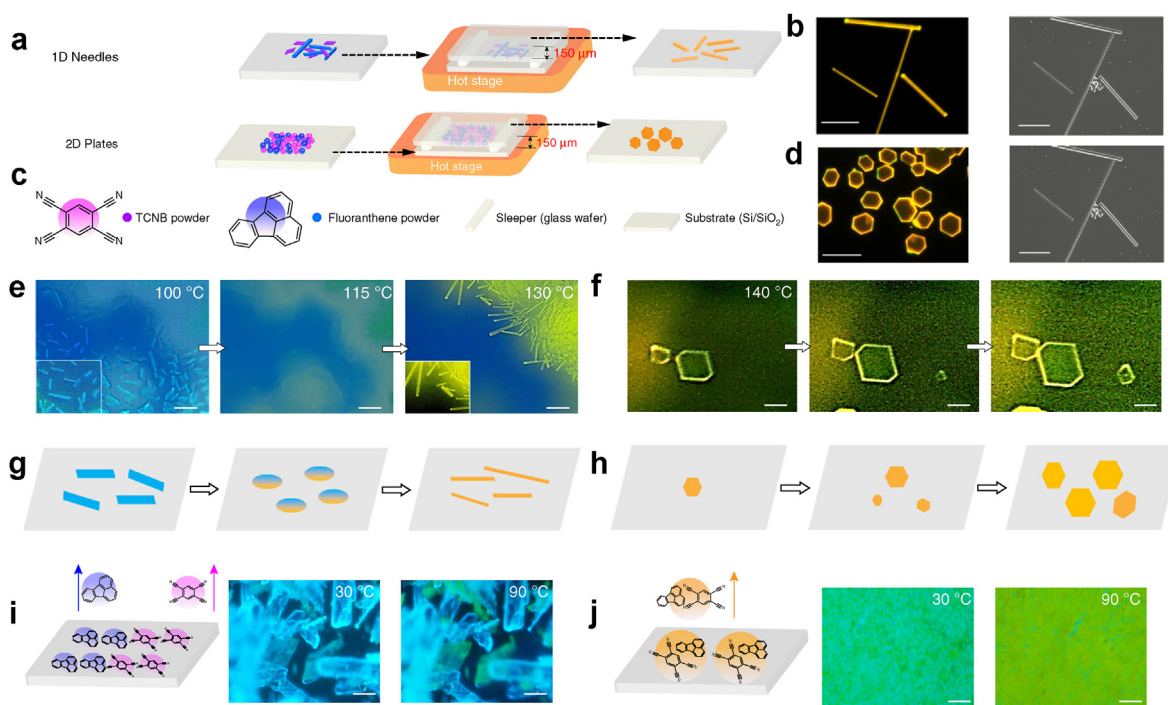


Fig. 10. MAS apparatus for the growth of 1D (a) and 2D (c) fluoranthene-TCNB cocrystals (FTCs). Fluorescence microscopy and SEM images of 1D (b) and 2D (d) FTCs, respectively. Real-time observation of crystal formation for 1D (e) and 2D (f) FTCs using a hot-stage fluorescence microscope on the top quartz substrates after heating. (g, i) Schematic representation of the crystallization mechanism for 1D FTCs in the MAS process. (h, j) Schematic representation of the crystallization mechanism for 2D FTCs in the MAS process. (Scale bar: 20 μm for e–f, 100 μm for others) [146]. Copyright 2019 Springer Nature.

high temperatures, which may cause compound degradation. Sublimation crystallization, on the other hand, has shown distinct advantages in screening and producing high-purity and high-quality cocrystals. This method allows for the direct transformation of a solid into vapor, which then re-deposits as a solid, effectively bypassing potential impurities or instabilities associated with the liquid phase.

For instance, Tao and colleagues [141] employed micro-distance sublimation on a hot-stage microscope (Fig. 8a) to grow a series of favipiravir cocrystals. Comparative analyses of diffusion flux (Fig. 8b and f), solubility (Fig. 8c), and tabletability (Fig. 8g) demonstrated that the cocrystals exhibited superior performance compared to the individual components. By increasing the size of the hot-stage and substrate, they successfully obtained high-quality single cocrystals (Fig. 8d–e). Further, by adjusting heating rates, growth distances, sublimation temperatures, and deposition conditions, Tao's group [141] subsequently synthesized favipiravir cocrystals with various coformers, including the cocrystals of FPV-isonicotinamide (FPV-INIA) (Fig. 8h and l), FPV-urea (Fig. 8i and m), FPV-tromethamine (FPV-Tro) (Fig. 8j and n), and FPV-nicotinamide (FPV-NIA) (Fig. 8k and o), underscoring sublimation as an effective method for cocrystal preparation. Similarly, Zhang et al. [142] utilized sublimation with nitrogen as a carrier gas to transfer the sublimated vapor into a crystal growth zone, where they synthesized a urea-succinic acid cocrystal, further highlighting sublimation's potential in cocrystal production.

To verify sublimation crystallization as a viable technique for cocrystal synthesis comparable to mechanochemical and solution methods (Fig. 9a), Carstens et al. [143] explored four benzoquinone-monohalogenated phenol cocrystals prepared *via* different crystallization techniques. While sublimation can theoretically produce all the cocrystal forms achievable by mechanochemistry and solution methods, slight polymorphic differences may arise in certain cases. However, by fine-tuning sublimation parameters, it is possible to rapidly screen for nearly all possible cocrystal forms, providing a powerful tool for comprehensive exploration of cocrystal systems.

Volkwyn et al. [144] further investigated the selective growth of cocrystals or organic salts *via* co-sublimation of acidic and basic small molecules (Fig. 9b). The formation of these compounds is highly influenced by the crystallization method employed. In solution crystallization, factors such as solvent polarity and the size of solvent molecules can affect the competition between hydrogen bonding and halogen bonding, thus impacting the crystal structure and composition. Lombard et al. [145] explored whether sublimation conditions, such as temperature, sublimation time, and the acid-base molar ratio, would influence the formation of salts or cocrystals, as well as the crystalline forms obtained. They found that longer sublimation times increased the quantity of the sublimated material but also favored the formation of organic salts. For the succinic acid (SA) and hexamethylenetetramine (HMT) system, reducing the sublimation time resulted in complete co-deposition of the material as co-crystals, while lower sublimation temperatures (higher than the sublimation point of any individual component) produce only co-crystals and do not form organic salts (Fig. 9c).

In further advancements, Ye et al. [146] achieved 1D and 2D morphological control of cocrystal materials based on polycyclic aromatic hydrocarbons through a micro-distance sublimation method (Fig. 10a–d). Their study focused on the growth of fluoranthene-TCNB (TCNB: 1,2,4,5-tetracyanobenzene) cocrystals, where fluorescence detection was employed to monitor the crystal growth under varying experimental conditions. When the unground material was sublimated at 130 °C, the lower melting component sublimated first and recorded on the substrate. As the temperature increased, fluoranthene melted into droplets on the substrate, and TCNB sublimated and reacted with fluoranthene molecules, crystallizing as 1D needle-like crystals within the droplets (Fig. 10i), and the growth process illustrated in Fig. 10e and g. After thoroughly grinding the initial material, co-sublimation at 140 °C for 30 min yielded 2D plate-like crystals (Fig. 10j), and the growth process depicted in Fig. 10f and h. Fluorescence microscopy revealed that the 1D crystals had lengths ranging from 10 to 400 μm, widths of

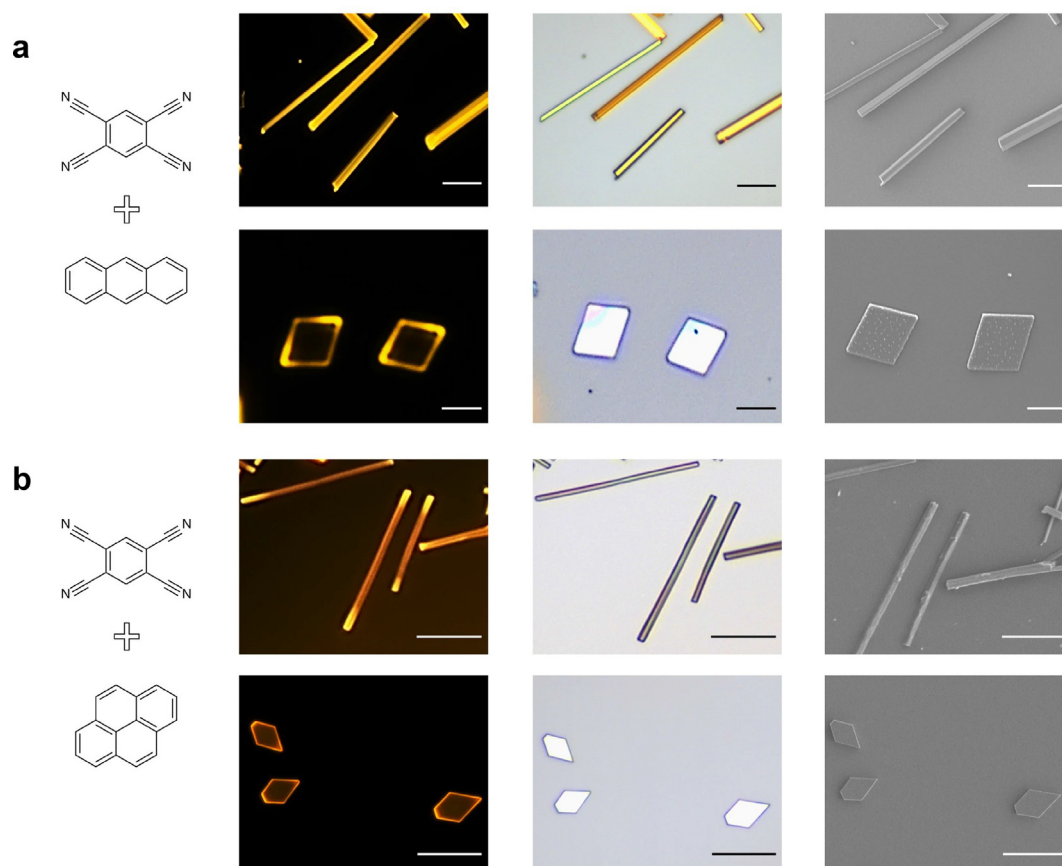


Fig. 11. (a) Fluorescence microscope, optical microscope, and SEM images of fluoranthene-TCNB cocrystals, showing the as-grown morphology of 1D and 2D structures. (Scale bar: 5 μm). (b) Corresponding images of pyrene-TCNB cocrystals, illustrating the growth characteristics in 1D and 2D configurations (Scale bar: 20 μm) [146]. Copyright 2019 Springer Nature.

approximately 5 μm , and thicknesses of 1–2 μm . The optical propagation loss measurements revealed that the 1D crystal had a loss coefficient of 0.0181 dB/ μm , while the 2D crystal showed lower coefficients of 0.0084, 0.0139, and 0.0147 dB/ μm on the (001), (012), and (01–2) planes, respectively. These values were notably lower than those of other organic crystals, such as Urea (0.01–0.1 dB/ μm), L-glutamic acid (0.05–0.2 dB/ μm), 1,5-diaminonaphthalene (DAN) (0.02–0.1 dB/ μm), aminonaphthol (ANP) (0.05–0.3 dB/ μm), and peptide nucleic acid (PNA) (0.01–0.05 dB/ μm).

To further verify the versatility of sublimation in preparing cocrystals, Ye et al. [146] applied the micro-distance sublimation method to the fluoranthene-TCNB, pyrene-TCNB, and pyrene-TIFB systems. By adjusting the growth conditions, they successfully controlled the transformation from 1D to 2D crystal morphologies. Consistent results were obtained through repeated sublimation experiments on fluoranthene-TCNB and pyrene-TCNB cocrystals, reinforcing the potential of the micro-distance sublimation method for the controlled fabrication of 1D and 2D organic materials (Fig. 11). Overall, sublimation crystallization technology has demonstrated its effectiveness in controlling the synthesis of cocrystals by regulating sublimation rates, vapor distribution, and temperature gradients. These findings not only broaden the application of sublimation crystallization in cocrystal preparation but also offer valuable technical guidance and theoretical support for the design and synthesis of advanced materials.

3.5. Synthesis of thin film

Thin film materials are integral to modern technology and widely used in optoelectronic devices, solar cells, and semiconductor

applications [147–149]. However, their fabrication processes often present significant challenges, particularly in achieving uniformity, crystallinity, and defect minimization [150]. Traditional methods, such as chemical vapor deposition, sputtering, and thermal evaporation, enable large-area film production but often fall short in terms of crystallization quality, precise stoichiometry, and film stability [151]. In comparison, sublimation offers distinct advantages due to its low contamination risk and ability to achieve high purity. By precisely controlling sublimation temperature and vapor conditions, this method allows for the fine-tuning of the crystal structure and physical properties of thin films, providing a reliable pathway for the fabrication of high-performance materials.

Metal halide perovskites (PVK) have garnered significant attention due to their high absorption coefficients, positioning them as promising candidates for next-generation solar cells [152]. Over the past decade, advancements in material optimization and fabrication techniques have pushed the power conversion efficiency (PCE) of PVK solar cells beyond 25% [153]. A critical factor in achieving such high PCE is the precise cation-halide mixing in the PVK structure. Compared to conventional solution methods, which suffer from poor reproducibility and unintended contamination, thin films prepared through sublimation and sequential deposition have demonstrated superior performance. For instance, Ngqoloda et al. [154] fabricated PVK thin films using sequential deposition, achieving a notable PCE of 21.32%, thereby validating the effectiveness of this method. Similarly, Yan et al. [155] employed sequential sublimation, deposition, and annealing to enhance diffusion and uniformity in $\text{Cs}_{0.15}\text{FA}_{0.85}\text{PbI}_{2.85}\text{Br}_{0.15}$ thin films (Fig. 12a). Using sublimation, materials were deposited via single-cycle (SCD) and multi-cycle deposition (MCD) methods onto glass substrates. XRD analysis revealed that films with optimized thicknesses (100 nm and 250 nm for SCD,

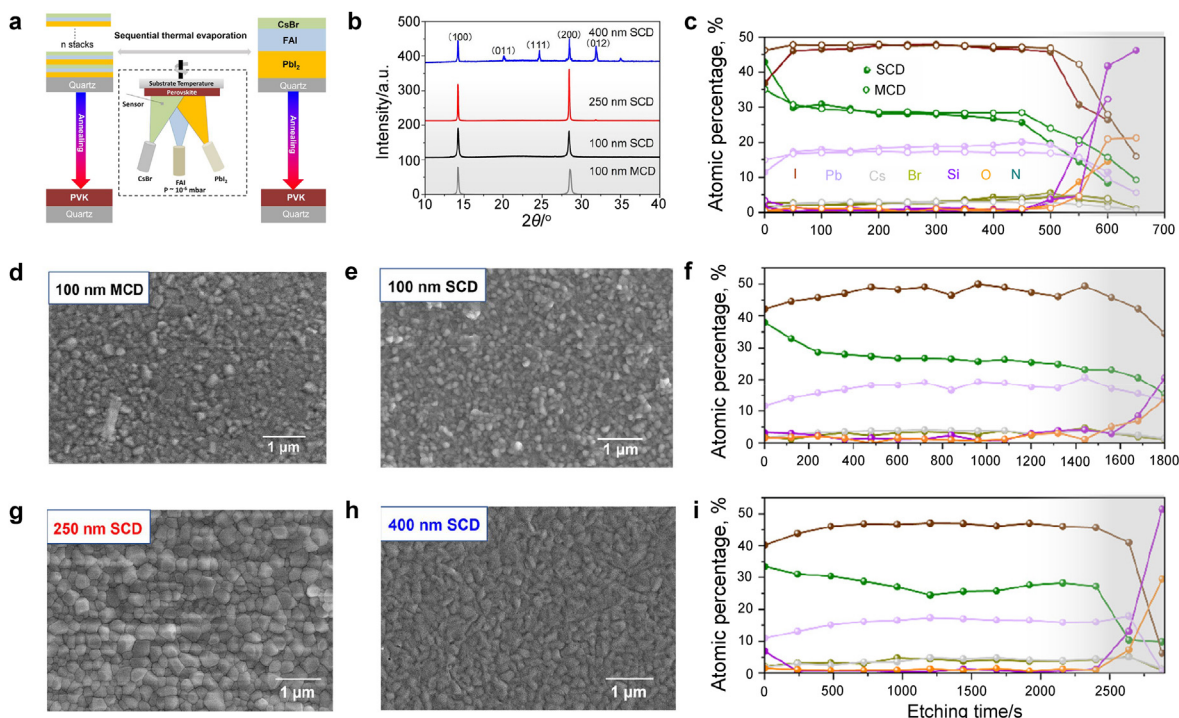


Fig. 12. (a) Schematic illustration of the apparatus used for single-cycle and multi-cycle sequential sublimation. (b) XRD patterns of MCD precursor films (gray) and SCD films, showing crystalline differences. (c) Elemental composition analysis of a 100 nm MCD film. (d) SEM image of a 100 nm MCD film, depicting surface morphology. (e) SEM image of a 100 nm SCD film, illustrating surface characteristics. (f) Elemental composition analysis of a 100 nm SCD film. (g) SEM image of a 250 nm SCD film, displaying surface features. (h) SEM image of a 400 nm SCD film, showing morphological details. (i) Elemental composition analysis of a 250 nm SCD film [155]. Copyright 2023 American Chemical Society.

100 nm for MCD) exhibited well-aligned crystal orientation, while films thicker than 400 nm (SCD) showed disorder (Fig. 12b). SEM observations demonstrated that grain size was slightly larger in multi-cycle deposition (130 nm) compared to single-cycle (100 nm), with increased nucleation sites contributing to this difference (Fig. 12d and e). Grain growth continued as film thickness increased from 100 nm to 250 nm (Fig. 12g), but at 400 nm, grains exhibited horizontal elongation and irregular orientation (Fig. 12h). XPS depth atomic percentage profile of PVK films with different thicknesses indicated uniformity in 100 nm films regardless of deposition technique (Fig. 12c and f), while 250 nm SCD films also maintained compositional consistency (Fig. 12i). However, a shift in composition was detected in 400 nm films, suggesting an impact of greater thickness on uniformity. Optical analysis revealed that thicker films displayed enhanced absorbance in the 500–800 nm range, though the absorption coefficients were similar across different deposition methods.

The sublimation method has proven to be highly effective in the fabrication of thin films, as demonstrated by Sujith et al. [156] who successfully prepared CsPbBr₃ thin films. The findings revealed that the material could sublime completely at room temperature, depositing as single-crystalline films with excellent thermal stability. CsPbBr₃ thin films exhibit broad optical absorption, direct bandgap, narrow emission peaks, high quantum yield, and superior optical performance, positioning them as promising candidates for optoelectronic applications. Further research by Zhu et al. [157] confirmed the material's thermal stability, showing that CsPbBr₃ remains stable below 690 K. Additionally, Yang et al. [158] developed CsPbBr₃-based solar cells with an optical bandgap of 2.33 eV and demonstrated high visible-light absorption coefficients, coupled with impressive photostability.

In another domain, germanium selenide (GeSe) has emerged as a strong contender for thin film applications due to its high environmental stability, layered structure, and tunable bandgap [159]. Despite its potential, current fabrication methods, such as vacuum laser thinning and

plasma thinning, suffer from inefficiencies and high equipment demands. GeSe's crystal structure with one short Ge-Se bond and two long Ge-Se bonds (Fig. 13a and b) and high vapor pressure (Fig. 13c) at temperatures below the melting point make it ideal for sublime-based 2D film fabrication, which can maintain the integrity of short bonds by breaking weaker long bonds to achieve efficient sublimation of GeSe (Fig. 13d). Xue et al. [160] used microspacing sublimation (setup shown in Fig. 13e) to create compact GeSe films. These films, when applied to solar cells, showed a photovoltaic effect without hysteresis between forward and reverse scans. The thin film displays a tightly packed structure, with its XRD patterns indicating a high level of crystallographic order and uniformity (Fig. 13f-i). External quantum efficiency (EQE) peaked at 540–700 nm but declined at other wavelengths, likely due to interface defects and film orientation. Although there is a gap between the EQE value of GeSe solar cells and that of high-efficiency CZTSSe solar cells, possibly due to differences in the GeSe/CdS heterojunction, the initial device performance and stability of GeSe solar cells demonstrate their great potential for thin-film solar cell applications (Fig. 13j-m). Further buffer layer optimization could improve performance. Wang et al. [161] further studied GeSe film growth on Si/SiO₂ substrates via vacuum sublimation, showing temperature-dependent morphology changes. At 80 °C, films formed as ultrathin layers, transitioning to thicker stripe-like structures from 120 °C to 150 °C, with optical properties shifting from blue to silver. At higher temperatures (200–250 °C), clustering reduced uniformity. Thinning a 175 nm GeSe film at 300 °C improved electrical properties by introducing Se vacancies, confirmed by SEM and energy dispersive spectroscopy (EDS) analysis. I–V measurements indicated a transition from linear to non-linear behavior as thinning progressed, showing increased resistance and vacancy effects, significantly lowering current compared to the original film.

By optimizing the sublimation temperature, the thinning of GeSe films has been shown to achieve maximum photoluminescence intensity and preserve overall crystal integrity. The thinned GeSe films exhibit an

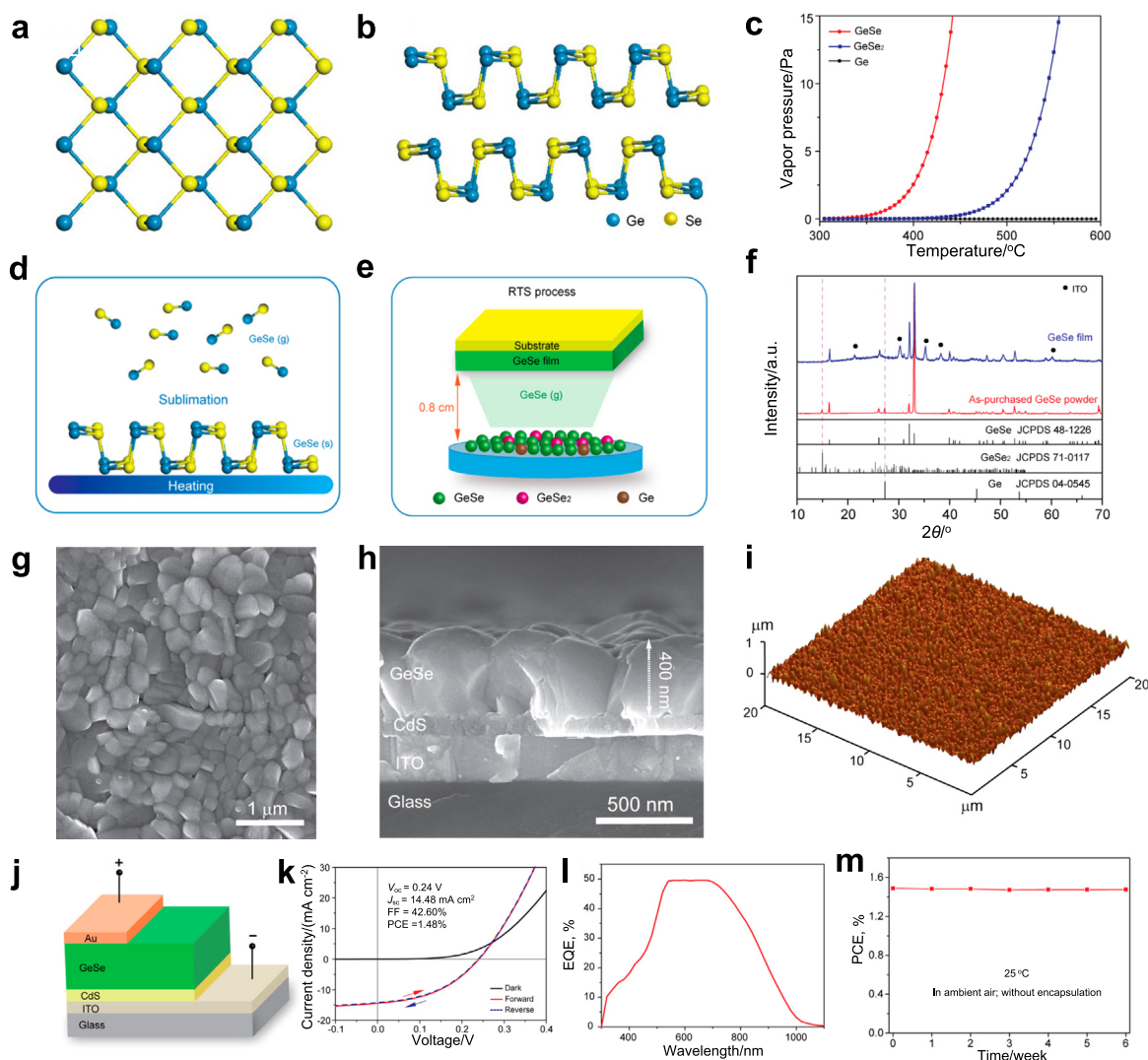


Fig. 13. (a) Top view of the orthorhombic crystal structure of GeSe. (b) Side view of the orthorhombic crystal structure of GeSe. (c) Saturated vapor pressure of GeSe as a function of temperature. (d) Schematic representation of the GeSe sublimation process. (e) Diagram of the sublimation apparatus. (f) XRD patterns of the raw material before and after sublimation, and the resulting GeSe thin films. (g) Top view SEM image of the GeSe film deposited on the CdS layer. (h) Cross-sectional SEM image of the GeSe thin film. (i) Atomic force microscopy (AFM) image of the GeSe thin film. (j) Schematic illustration of the GeSe-based solar cell. (k) Forward and reverse J-V curves of the GeSe solar cell under dark conditions and 100 mW cm^{-2} illumination. (l) External quantum efficiency (EQE) spectrum of the GeSe solar cell. (m) Stability of the GeSe solar cell at room temperature [160]. Copyright 2017 American Chemical Society.

orthorhombic lattice structure with direct bandgap properties. Furthermore, the electrical conductivity of bipolar devices fabricated from these thinned films demonstrated the preservation of crystal integrity after sublimation. This sublimation thinning method is not only efficient and straightforward but also presents a scalable solution for producing high-quality 2D materials. These findings further substantiate the application potential of sublimation in thin-film material production, highlighting its significant role in improving film quality, minimizing defects, and optimizing material performance.

3.6. Performance of transistor

Transistors are primarily classified into bipolar junction transistors (BJTs) and field-effect transistors (FETs), with the latter including metal-oxide-semiconductor field-effect transistors (MOSFETs) and junction field-effect transistors (JFETs), both of which play critical roles in modern electronic devices [162,163]. Various fabrication techniques are employed to manufacture transistors, such as chemical vapor deposition, molecular beam epitaxy, and laser annealing [164,165]. While each

method has its unique characteristics, sublimation has gained attention for its specific advantages. In the production of p-n junction devices and complementary logic circuits, electron-transporting (n-type) and hole-transporting (p-type) organic materials are commonly used [166]. Positive-type organic field-effect transistors (OFETs), such as those based on ethylene and oligothiophenes, have demonstrated superior performance [167,168]. However, high-performance n-type OFETs remain less common due to challenges in achieving reliable electron injection.

p-type compounds with negatively charged groups have been shown to enable high-performance n-type OFETs. For example, perfluoropentacene, through full substitution of pentacene, adjusts its polarity for n-type OFETs, but challenges such as disproportionation during synthesis can reduce purity and affect device performance. Hence, more stable materials are needed. Kikuzawa et al. [169] found that 2,5,8,11,14,17-hexafluoro-hexa-peri-hexabenzocoronene (6F-HBC, Fig. 14a) exhibits high chemical stability. The rigidity of its core and the planar conjugation of PI promote self-assembly, resulting in materials with high charge carrier mobility. The team synthesized 6F-HBC by fluorine substitution, though with slightly reduced stability compared to HBC.

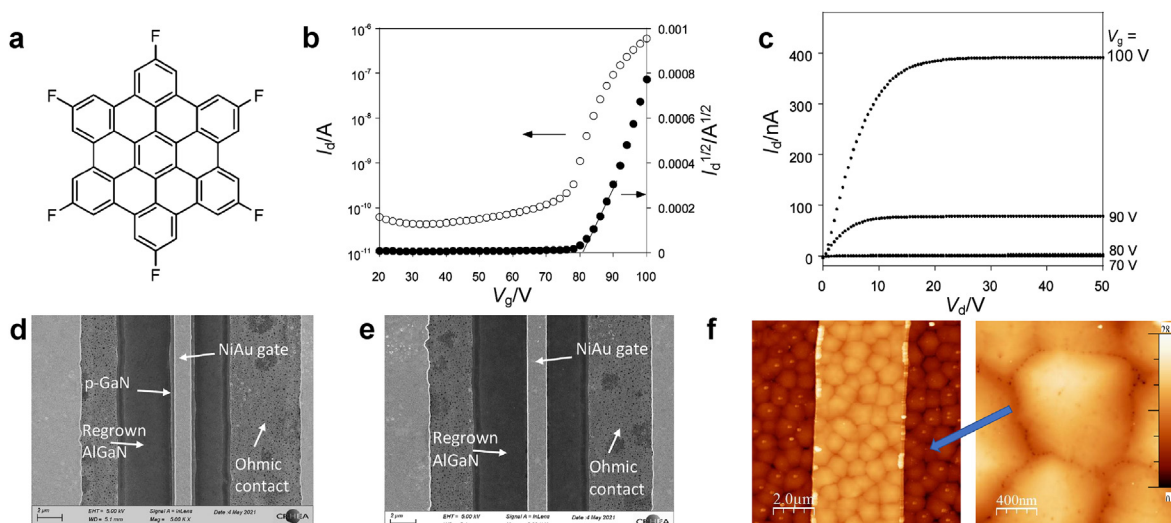


Fig. 14. (a) Molecular structure of 6F-HBC. (b) Transfer characteristics of the OFET with 6F-HBC as the active layer. (c) Output characteristics of the OFET with 6F-HBC as the active layer [169]. Copyright 2007 American Chemical Society. SEM images of an (d) E-type transistor and (e) D-type transistor. (f) AFM images of the gate area after p-GaN evaporation, AlGaIn regrowth, and SiO₂ mask removal, as well as the AFM view of the regrown AlGaIn surface [170]. Copyright 2022 Elsevier.

Comparing the spectra of HBC and 6F-HBC, they observed HOMO levels of 5.4 eV and 5.9 eV, respectively, with fluorine lowering the LUMO level, enhancing electron injection. OFET devices using 6F-HBC showed excellent n-type performance and high mobility (Fig. 14b–c).

Gallium nitride (GaN) is a prime candidate for high-power and high-frequency applications due to its high breakdown voltage, fast electron transport, and excellent thermal conductivity. Ngo et al. [170] used plasma-enhanced chemical vapor deposition (PECVD) on SiO₂ substrates (Fig. 14d–f), followed by selective evaporation and AlGaIn regrowth to fabricate transistors (Fig. 15a–c). Transistors mainly include D-mode transistor and E-mode transistor (Fig. 15d–e). The regrown structure significantly improved two-dimensional electron gas (2DEG) carrier concentration ($1.1 \times 10^{13}/\text{cm}^2$), reduced sheet resistance ($\sim 400 \Omega/\text{m}^2$), and increased electron mobility ($> 1100 \text{ cm}^2/\text{V}\cdot\text{s}$), outperforming previous designs without the 2 nm AlN layer (Fig. 15f). Transmission line method measurements showed a reduced ohmic contact resistance ($< 0.3 \Omega \text{ mm}$). In E-mode transistors (Fig. 15d), total resistance decreased to $\sim 8 \Omega \text{ mm}$ (Fig. 15g), and while thermal effects caused a slight drain current drop at high V_{ds} , the hysteresis was minimal (Fig. 15h–i). For D-mode transistors (Fig. 15j), the threshold voltage was -5.6 V , with the drain current reaching 500 mA/mm at $V_{\text{gs}} = 0 \text{ V}$ and over 700 mA/mm at $V_{\text{gs}} = +2 \text{ V}$, also showing minimal hysteresis (Fig. 15k–l). The regrown AlGaIn layer significantly enhanced DEG carrier concentration and reduced contact resistance, improving transistor performance (Fig. 15j).

Lin and colleagues [92] utilized the MAS technique (Fig. 16a) to overcome traditional vapor growth limitations by controlling substrate wettability, producing uniform single-crystal thin films. These films enabled efficient charge transport, ideal for high-performance integrated electronics. A transistor array was fabricated on the films using a top-contact and bottom-gate structure with 300 nm SiO_2 as the dielectric and 30 nm Au for the electrodes. All OFETs in the array operated successfully, achieving 100% yield. The transmission and output curves (Fig. 16b–d) demonstrated excellent linearity with typical p-type characteristics. The spatial mobility distribution (Fig. 16e) showed uniformity across the films, and the histogram (Fig. 16f) revealed an average mobility of $3.91 \text{ cm}^2/\text{V}\cdot\text{s}$, with a reliability range of 74.2%–99.5%. The highest measured mobility was $5.61 \text{ cm}^2/\text{V}\cdot\text{s}$, surpassing devices on SiO₂ substrates, with stable performance over four months.

Sublimation involves heating a solid material to directly convert it into gas, which then condenses at lower temperatures to form high-purity crystals. This process efficiently removes impurities, ensuring superior

crystal quality and material purity. Compared to traditional methods like chemical vapor deposition (CVD) and molecular beam epitaxy (MBE), sublimation offers significant advantages in terms of material purity and crystal quality. This is especially true for high-performance semiconductors such as GaN and SiC, where sublimation produces crystals with low defect densities. Although the growth rate of sublimation is relatively slow, its ability to yield high-purity and defect-free crystals makes it a key method for fabricating reliable and high-performance electronic devices. In summary, sublimation provides an efficient and high-quality approach for transistor fabrication, particularly for demanding semiconductor materials. As technology advances, sublimation will likely play an increasingly important role in future electronic device manufacturing.

4. Conclusions and future perspectives

Sublimation crystallization stands out as an efficient separation and purification technique, offering the advantage of producing high-purity products without the need for solvents. Its potential applications span materials science, pharmaceuticals, and electronics. By controlling the thermodynamic and kinetic conditions, sublimation crystallization facilitates crystal growth and phase transitions while avoiding solvent-related contamination common in traditional methods. However, despite its clear advantages, some limitations need to be addressed to fully unlock its potential.

One key limitation is that the effectiveness of sublimation crystallization depends on the properties of the materials. Not all substances are suitable for this method, as materials with low volatility or poor thermal stability may decompose before reaching the sublimation temperature, making the process infeasible. For heat-sensitive materials, developing alternative approaches such as protective atmospheres during sublimation could help prevent decomposition. Additionally, in complex multi-component systems, achieving high-quality and defect-free crystals can be challenging. Variations in temperature gradients or inconsistencies in the sublimation environment can result in defects or the formation of unwanted polymorphs, which may affect the material's performance. Optimizing the growth conditions to control crystal morphology and ensure phase purity remains an area requiring further investigation.

While sublimation crystallization performs well in laboratory settings, scaling up industrial applications presents technical challenges. It is difficult to maintain uniform temperature and pressure across large

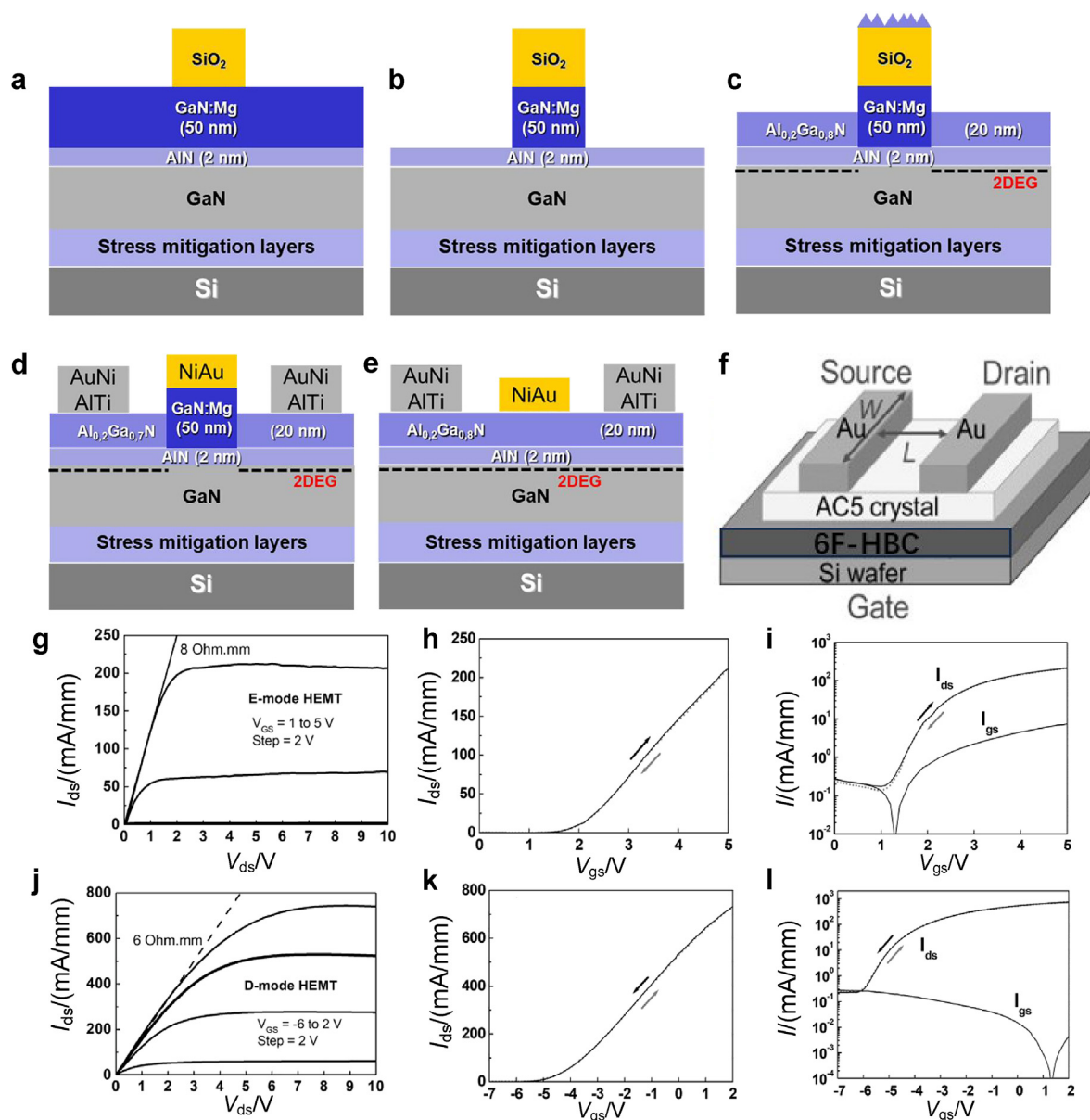


Fig. 15. (a) Schematic cross-section showing the initial epitaxial layers covered with SiO_2 . (b) Selective evaporation process of the p-GaN cap layer. (c) Regrowth of the AlGaIn layer. (d) Illustration of an E-mode transistor. (e) Illustration of a D-mode transistor. (f) Schematic structure of an OFET. (g) DC output characteristics of a $1.5\ \mu\text{m}$ gate E-mode transistor. (h) Hysteresis transfer characteristics at $V_{\text{ds}} = 7\ \text{V}$, showing I_{ds} for positive (solid line) and negative (dashed line) sweeps of gate bias. (i) Logarithmic plot of the drain current I_{ds} and gate current I_{gs} recorded under identical conditions. (j) DC output characteristics of a $1.5\ \mu\text{m}$ gate D-mode transistor. (k) Hysteresis transfer characteristics at $V_{\text{ds}} = 7\ \text{V}$, with I_{ds} for positive (solid line) and negative (dashed line) sweeps of gate bias. (l) Logarithmic plot of the drain current I_{ds} and gate current I_{gs} under the same testing conditions [170]. Copyright 2022 Elsevier.

areas or in large reactors, which is essential for consistent crystal quality. Advanced equipment and process optimization will be needed to address these scale-up issues. Moreover, while the process eliminates solvent use, energy consumption is still a concern, especially for materials with high sublimation temperatures. Efforts to develop low-energy sublimation techniques and enhance thermal efficiency are crucial to advancing the sustainability of this method.

Future research directions are likely to focus on hybrid approaches that combine sublimation crystallization with other techniques, such as vapor deposition, chemical vapor transport, or mechanochemical synthesis, to overcome existing limitations. These hybrid methods could improve control over crystal growth dynamics, facilitate the development of novel materials, and broaden the applicability of sublimation. Progress in real-time monitoring and feedback control technologies will also play a key role in optimizing process parameters to enhance product

quality and reduce defects. Additionally, sustainable practices should be prioritized, including low-energy sublimation technologies, improved heat recovery systems, and the use of renewable energy to reduce the carbon footprint. Crystal engineering efforts that aim to optimize intermolecular forces and lattice energy for better control of solid-state phase transitions represent another promising research avenue.

Moreover, although there are no current applications of sublimation crystallization in chiral separations, the technique's molecular-level precision offers promising prospects for future applications. We propose that future research could explore the integration of sublimation crystallization with chiral recognition molecules and optimize crystallization conditions to improve the separation efficiency of chiral isomers.

In summary, sublimation crystallization offers unique advantages in producing high-purity and environmentally friendly materials, but challenges related to energy consumption, industrial scale-up, and

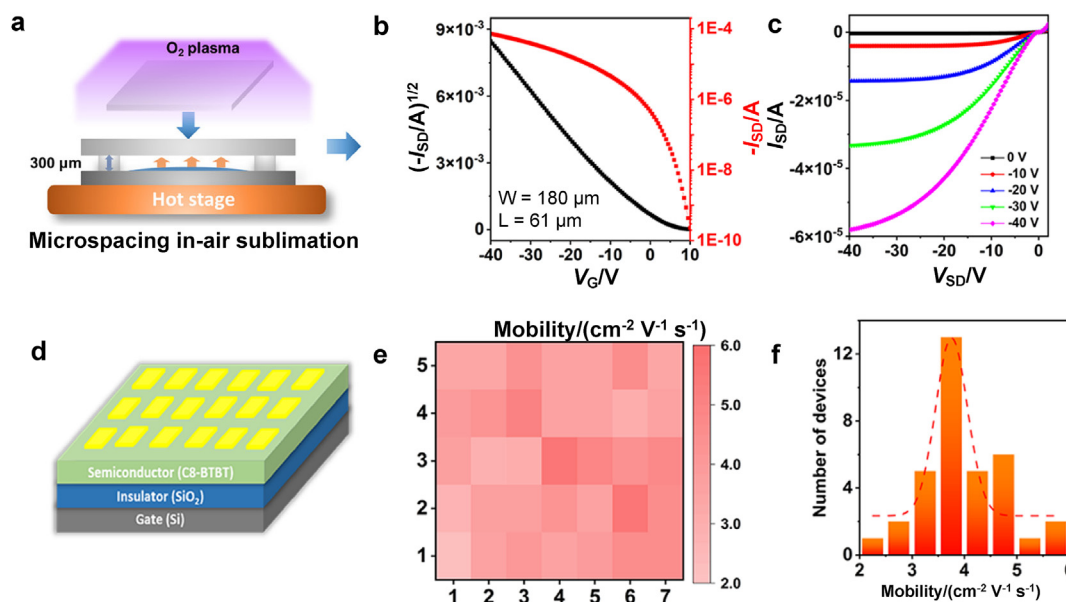


Fig. 16. (a) Schematic illustration of the microspacing sublimation apparatus. (b) Transfer characteristics of a representative OFET device built on a C8-BTBT (BTBT: benzo[b]thieno[3,2-b]thiophene) single-crystalline film, measured at $V_{SD} = -40$ V. (c) Output characteristics of the same OFET device. (d) Schematic depiction of the device array layout. (e) Color map showing the carrier mobility (μ) values across a 5×7 array of devices (8 mm^2 total area). (f) Distribution histogram of carrier mobility values for the 35 devices within the array [92]. Copyright 2022 American Chemical Society.

process control need to be addressed. As process parameters are refined and technological advancements continue, sublimation crystallization is expected to play an increasingly significant role in the fields such as catalysis, energy storage, and the production of optoelectronic materials, establishing itself as a key technology for future high-purity material synthesis.

CRediT authorship contribution statement

Sai Wang: Writing – original draft, Methodology, Investigation. **Yuntian Xiao:** Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Yitian Su:** Methodology, Investigation. **Yongkang Liu:** Visualization, Validation, Supervision. **Shanshan Feng:** Writing – review & editing, Visualization. **Hongchen Cao:** Validation, Formal analysis. **Ling Zhou:** Supervision. **Qiuxiang Yin:** Writing – review & editing, Supervision, Software, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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